

SPEF File

Monday, September 21, 2026 10:39 AM

1 What is a .SPEF file ?

From Rout
to Accurat

1 Definition & Full Form



- **Full Form: Standard Parasitic Exchange Format**
- Defined by: **IEEE 1481**
- Type: **ASCII** parasitic exchange format
- Contains: Extracted resistance (R) and capacitance (C) of every routed wire
- Used by: STA tools (Tempus, PrimeTime, etc.) for accurate timing analysis

2 Why SPEF is needed after routing ?

- Before routing, the tool does not know the exact wire geometry, layer, width, length and neighboring wires.
- Hence it uses **wire load model (WLM)** to estimate RC (an educated guess).
- **After routing**, the exact physical information is available (real wire locations and geometry).
- **SPEF** provides the actual extracted parasitics based on the routed layout, so STA can calculate accurate delays for signoff.

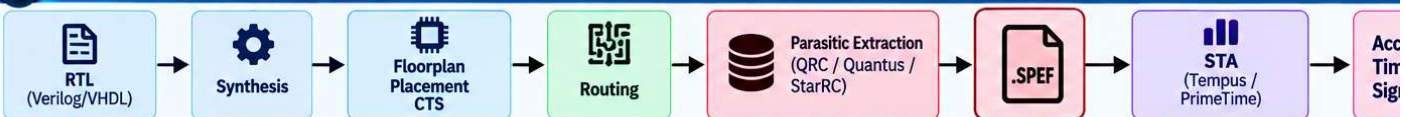
"Without SPEF: timing is an educated guess.
With SPEF: timing is a measurement of reality."

3 IEEE 1481 and SPEF



- SPEF is standardized by **IEEE** (Standard for Parasitic Exchange Format)
- It is an open, industry-standard supported by major EDA tools (Cadence, Synopsys, etc.).
- Being a standard format, a SP by one tool (e.g., QRC/Quantus) by another tool (e.g., Tempus/PrimeTime).

4 From RTL to STA – Where SPEF Fits ?



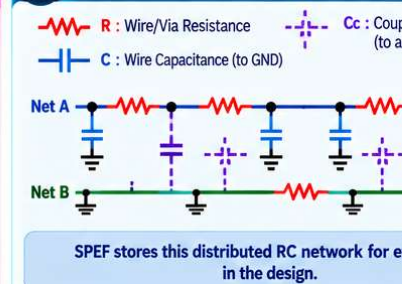
5 What information does SPEF contain ?

Category	Information in SPEF
Parasitic Data	Resistance (R) and Capacitance (C) for every routed net (distributed RC network)
Net Structure	Net names, connectivity, RC nodes, port information
Capacitance Types	Ground capacitance and coupling capacitance (between adjacent wires)
Resistance Types	Wire segment resistance and via resistance
Units	Time, capacitance, resistance and inductance units (e.g., T_UNIT, C_UNIT, R_UNIT)
Hierarchy Mapping	Name mapping (compressed net/instance names)
Design Metadata	Design name, tool, version, date, etc.

6 What does SPEF NOT contain ?

Category	Not present in SPEF
Cell Timing	No cell delay, no library timing information (that comes from .lib files)
Logical Function	No RTL/logic behavior
Layout Geometry	No exact physical coordinates, shapes, polygons or layer information (that is in DEF/GDS)
Power Information	No power intent, voltage domains (that comes from UPF)
Clock Constraints	No timing constraints (that comes from SDC)
Library Data	No standard cell library or PVT information
Design Rules	No DRC/LVS information

7 Simple Example – What SPEF Represents



Key Takeaway:

"SPEF = Actual parasitics of the routed design, standardized (IEEE 1481), used by STA tools for accurate timing signoff. It **does not contain** logical, library or layout geometry information."

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2 Why do we need SPEF after Routing ?

After routing, the real wire geometry is known. SPEF captures the actual parasitics (R and C) for accurate timing analysis.

Real
Real
Res

1 Pre-route RC Estimation (Wire Load Model)

- Before routing, the exact physical layout of wires is unknown.
- Tools use **Wire Load Model (WLM)** to estimate resistance and capacitance based on net fanout, cell type and technology.
- It is only an educated guess.
- Used during synthesis and early timing analysis.

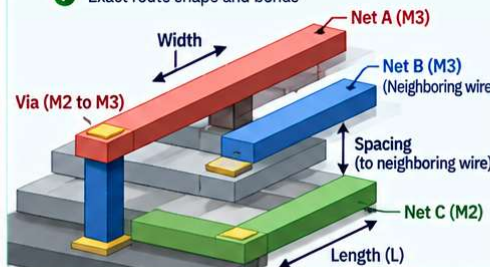


Example (WLM Estimation)

Net fanout	3
Estimated resistance (R)	~ 5 Ω
Estimated capacitance (C)	~ 20 fF
Model type	Wire Load Model (technology based)
Accuracy	Low (only for early stage)

2 After Routing – Real Geometry is Known

- After routing, every wire is physically present in the layout.
- We know the exact geometry and environment for each wire:
 - ✓ Metal layer
 - ✓ Width
 - ✓ Length
 - ✓ Via locations
 - ✓ Neighboring wires (for coupling capacitance)
 - ✓ Exact route shape and bends



3 Pre-route vs Post-route Comparison

Aspect	Pre-route (WLM)	Post-route
Wire geometry	Unknown	Known
Metal layer	Estimated	Actual layer (e.g., M3)
Width	Not known	Exact
Length	Estimated from fanout	Exact route length
Via locations	Not known	Exact via locations
Neighboring wires	Not considered	Known (for coupling capacitance)
Parasitic R and C	Estimated (lumped)	Extracted (distributed)
Tool used	Synthesis / early STA	QRC/Quantus (after routing)
Accuracy	Low	High (signoff)

4 Why Post-route Timing is More Accurate ?

- Uses actual extracted parasitics (R and C) from the routed layout.
- Considers real wire resistance, ground capacitance and coupling capacitance.

5 Impact on Timing

Parameter	Pre-route (WLM)	Post-route (SPEF)
Net delay	Estimated	Actual (extracted)
Path delay	Less accurate	More accurate

6 Key Takeaway



"SPEF is needed after routing because it provides the actual parasitics for accurate timing analysis."

- Captures impact of long wires, detours, vias and congested regions.
- Includes coupling effects from neighboring wires (important for SI).
- Eliminates the uncertainty of estimation, giving true delays.
- Used for signoff timing analysis (Tempus, PrimeTime, etc.).

Setup/Hold slack	Can be optimistic or pessimistic	Reliable for signoff
Crosstalk/SI	Not considered	Captured (coupling C)
ECO / Optimization	Based on estimate	Based on real data
Signoff readiness	No	Yes

parasitic information (R and C) based on the real routed geometry, making the timing analysis **and signoff-ready.**"



Remember:

"Pre-route is an **estimate**, Post-route (SPEF) is the **reality**."



Accurate Parasitics Lead to Reliable Timing

3

What is Parasitic R and C in a Routed Wire ?

A real interconnect has resistance and capacitance, which impact signal delay and signal integrity.

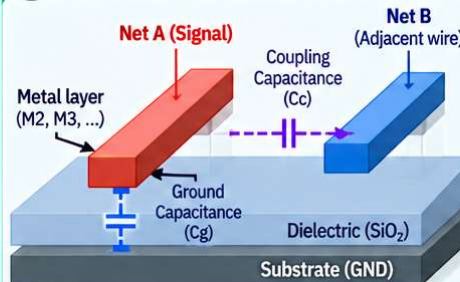
Real Wires Have Parasitics They Ideal

1 Why a Real Wire is not an Ideal Conductor ?

- A real wire is made of metal with finite resistivity, so it has **resistance (R)**.
- It is separated from the substrate and nearby wires by dielectric material, so it forms **capacitance (C)**.
- At high frequencies, electric and magnetic fields also introduce additional parasitic effects.
- Therefore, a wire does not behave like a perfect wire (0 Ω and 0 capacitance).
- These parasitics become significant as technology node shrinks and frequency increases.

A real wire = Resistor + Capacitor(s) (not an ideal conductor)

2 Types of Parasitic Components



Each wire has:

- Series resistance (R) along its length
- Capacitance to ground (Cg)
- Coupling capacitance to nearby wires (Cc)

3 Wire Resistance (R)

- Due to finite resistivity of metal, a wire has resistance.
- Resistance increases with **length** and **decreases** with higher conductivity metal.

$$R = \rho \times \frac{L}{A} = R_s \times \left(\frac{L}{W} \right)$$

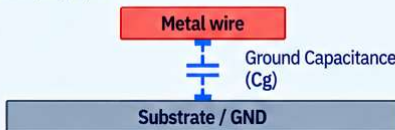
ρ : resistivity
L : length of wire
A : cross-sectional area
 R_s : sheet resistance
W : width

Example:

- Metal M2 with sheet resistance $R_s = 0.05 \Omega/\square$
- Length = 10 μm , Width = 0.1 μm $\rightarrow$ Number of squares = 100
- $R = 0.05 \times 100 = 5 \Omega$

4 Ground Capacitance (Cg)

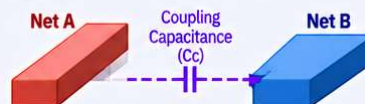
- Capacitance between the wire and the reference ground (substrate or ground plane) through the dielectric.
- Also called area capacitance and fringe capacitance.
- Caused by the electric field between the wire and the underlying layers.



Cg depends on wire width, thickness, dielectric constant and distance to ground.

5 Coupling Capacitance (Cc)

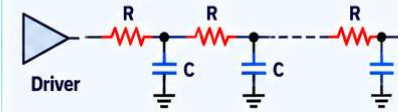
- Capacitance between two adjacent parallel wires.
- Occurs due to the electric field between nearby wires.
- Depends on spacing, length of parallel run, and dielectric properties.
- Responsible for crosstalk and noise.



Cc increases with longer parallel run length and smaller spacing between wires.

6 How R and C Affect Signal Delay

- Wire resistance (R) and capacitance (C) form an RC network (distributed along the wire).
- They slow down the signal transition, increasing propagation delay.
- Higher R and C $\rightarrow$ larger RC time constant $\rightarrow$ more delay.
- Large coupling capacitance can also cause crosstalk leading to timing failures.



Wire delay $\approx R_{\text{total}} \times C_{\text{total}}$ (simplified)
In reality, it is a **distributed RC network** solved by SPICE.



Key Takeaway:

"Every routed wire has parasitic R and C (ground and coupling). These parasitics slow down the signal and must be accurately extracted (SPEF) for signoff timing."



Accurate Parasitics = Accurate Timing = Reliable Design

4

What is the difference between WLM/estimated RC and SPEF/extracted RC ?

WLM is an early estimate based on models, while SPEF is the actual extracted parasitics from the routed design.

Diff. Approach Saves Accuracy

1 What is Wire Load Model (WLM) ?

- A pre-route estimation technique used during synthesis and early design stages.
- Estimates wire resistance (R) and capacitance (C) based on net fanout, cell type and technology.
- Uses predefined tables/models (wire load models) provided by the PDK.
- Does not know the actual routing, wire geometry or congestion.
- Provides a **single lumped** R and C per net.
- Used for early timing analysis and optimization.

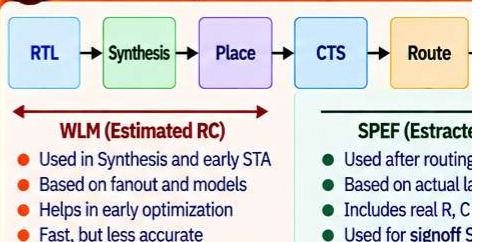
WLM = Educated Guess

2 What is SPEF (Extracted RC) ?

- SPEF (Standard Parasitic Exchange Format) contains the actual resistance (R) and capacitance (C) extracted from the routed layout.
- Generated by extraction tools (QRC/Quantus/StarRC) after routing.
- Contains **distributed RC network** (multiple R and C segments per net).
- Considers real wire geometry: metal layer, width, length, via locations and neighboring wires (coupling).
- Used for post-route STA (signoff timing).

SPEF = Measurement of Reality

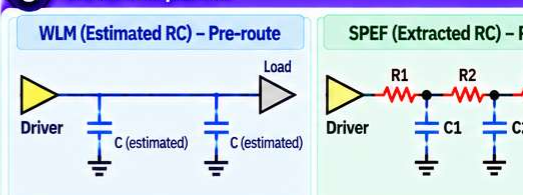
3 When are they used ? (Design Flow View)



4 Key Differences

Aspect	WLM / Estimated RC	SPEF / Extracted RC
Stage	Pre-route (Synthesis / Early STA)	Post-route (Signoff STA)
Source	Predefined wire load models (PDK)	Extracted from actual routed layout (QRC/Quantus/StarRC)
Information used	Fanout, cell type, wire load tables	Actual wire geometry, metal layer, width, length, vias, neighbors
RC value	Estimated (single lumped R and C per net)	Actual extracted (distributed R and C network)
Coupling capacitance	Not considered	Included (ground + coupling capacitance)
Accuracy	Low (only an estimate)	High (signoff quality)

5 Visual Comparison



File/Input	WLM file (lib/tech based)	SPEF file (IEEE 1481 format)
Tool usage	Synthesis tools (DC, Genus, etc.)	STA tools (Tempus, PrimeTime, etc.)
Purpose	Early timing and optimization	Final timing analysis and signoff

Single lumped R and C per net
(based on fanout)

Distributed R and C
(actual values from I)

6 Why SPEF Replaces Estimation during Post-route STA ?

- After routing, the exact wire geometry and parasitics are known.
- Timing analysis needs real RC values for accuracy.
- SPEF includes coupling capacitance, congestion effects, detours and vias.
- WLM is no longer valid because the actual route may be very different from the model.
- Signoff timing (e.g., hold/setup, crosstalk, ζ) requires extracted parasitics.

As soon as the design is routed,
we stop guessing and start using
the real extracted parasitics (SPEF)
for accurate and reliable timing analysis



Key Takeaway:

"WLM gives an **estimate** to start the journey,
but **SPEF** gives the **real parasitics** to reach the destination (signoff)."



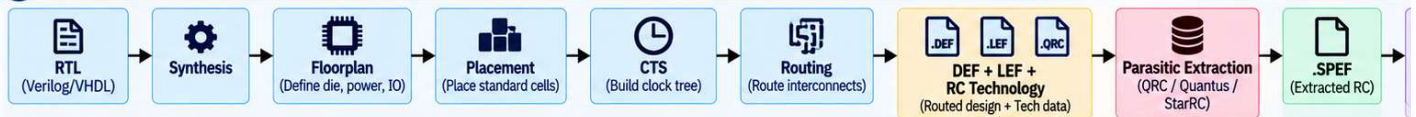
Estimation is u
Extraction is e

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How is SPEF generated from a routed design ?

SPEF is generated after routing by **extracting parasitic R/C** from the routed physical database using layout and RC technology information, then consumed by STA.

1 Complete Flow – From RTL to STA



2 Where does extraction happen ?

- Extraction happens **after routing**, when the routed physical design is available.
- Parasitic extraction tools (Cadence Quantus/QRC or Synopsys StarRC) read the routed design and technology data to compute R and C.
- It is part of the signoff flow (post-route).

cadence
Quantus / QRC
(used with Innovus)

synopsys
StarRC
(used with ICC2)

3 Why must routing exist ?

- Before routing, exact wire geometry is unknown, so tools use wire load models (estimated RC).
- After routing, the real geometry is known:
 - Metal layer (M1, M2, ...)
 - Wire width and length
 - Via locations
 - Exact route shape and bends
 - Neighboring wires (for coupling)
- This information enables extraction of distributed R and C for accurate timing.

4 Role of DEF (Routed Design)

- Routed DEF contains the complete physical routing information.
- It provides the exact location of wires, vias, metal layers and net connectivity.
- Extraction tools use this routed information to build the interconnect model.

.DEF
(Routed)



5 Role of LEF (Technology Data)

- LEF provides physical abstracts required for extraction and physical design.
- Technology LEF:** defines routing layers, via definitions, design rules, layer thickness, spacing, etc.
- Cell/Macro LEF:** provides physical abstracts of standard cells, macros and blockages.

.LEF
Tech LEF
(layer/via info)

.LEF
Cell/Macro LEF
(cell abstracts)

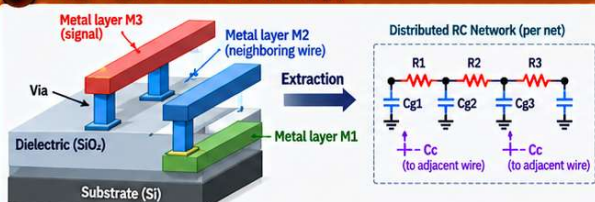
6 Role of RC Technology

- RC technology data (e.g., qrcTech) contains process and interconnect modeling information.
- Includes sheet resistance, capacitance parameters (area/fringe), via and layer-specific RC values.
- Used by extraction tool along with geometry to calculate parasitics for the geometry.

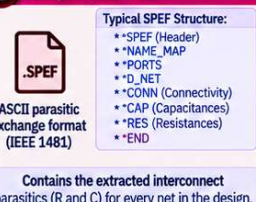
.qrcTech

Contains:
• Sheet resist
• Capacitance
• Via resistance
• Layer model

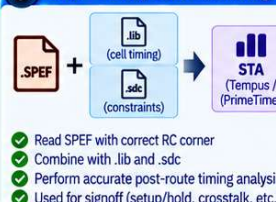
7 Inside Parasitic Extraction (Concept)



8 Output: SPEF File



9 SPEF → STA (Post-route Timing)



10 Pre-route vs Post-route

Aspect	Pre-route (WLM)	Post-route (SPEF)
Geometry	Unknown	Known
Method	Wire load model	Exact extraction
R/C value	Estimated (lumped)	Exact (distributed)
Coupling C	Not considered	Considered
Accuracy	Low (early stage)	High (signoff)
Used for	Synthesis, early STA	Final STA, signoff



Key Takeaway:

"SPEF is generated **after routing** using the routed physical design (DEF), technology data (LEF + RC tech), and a parasitic extraction tool. It provides the **extracted RC network** for accurate **post-route timing analysis**."

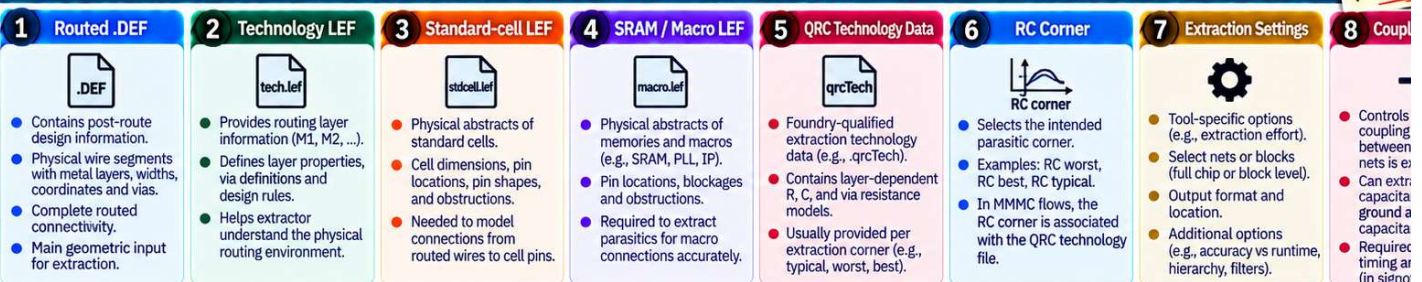


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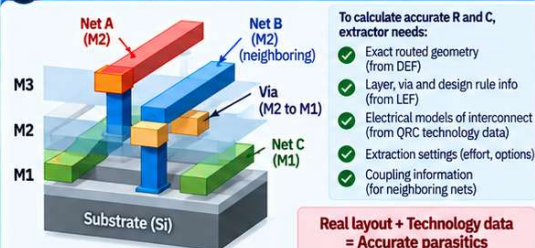
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What inputs are required for parasitic extraction ?

Parasitic extraction uses the routed layout geometry along with technology and tool settings to calculate the actual R and C of the design.



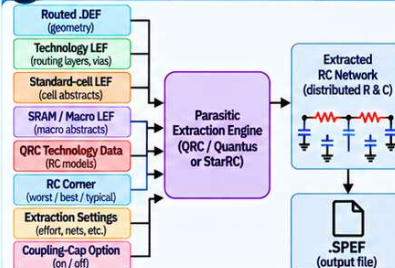
9 Why These Inputs Are Needed ?



- To calculate accurate R and C, extractor needs:
- Exact routed geometry (from DEF)
 - Layer, via and design rule info (from LEF)
 - Electrical models of interconnect (from QRC technology data)
 - Extraction settings (effort, options)
 - Coupling information (for neighboring nets)

Real layout + Technology data
= Accurate parasitics

10 Overall Flow – Inputs to Extraction Engine



11 Input Summary Table

Input	What it tells the extractor	Why it is needed
Routed .DEF	Post-route geometry, metal layers, wire segments, vias, connectivity	Gives exact physical data for RC calculation
Technology LEF	Routing layers, via definitions, layer properties, design rules	Helps interpret design correctly
Standard-cell LEF	Cell size, pin locations, pin shapes, obstructions	Needed to extract parasitics for standard cells
SRAM / Macro LEF	Macro abstracts, pins, obstructions	Required for macro parasitics
QRC Technology Data	Layer R, C, via resistance models (process/interconnect data)	Defines how parasitics are modeled
RC Corner	Selects corner (worst/best/typical)	Generates parasitics for desired corner
Extraction Settings	Effort, selected nets, output format, other tool options	Controls accuracy and scope
Coupling-Cap Option	Whether to extract coupling capacitance between adjacent nets	Required for accurate timing analysis

12 Tools in Industry (Examples)



Note:
Exact input file set and options may vary by tool, version and methodology.
This flow shows basic

13 Coupling vs Ground Capacitance (Visual)



(used with Innovus)

(used with ICC2)

• Supports industry standard formats (e.g., SPEF)

The color between wires represents a typical LEF/DEF-based digital extraction flow.

Ground Capacitance (C_g) (to substrate)

(between adjacent wires)

is important for and signoff ti

**Key Takeaway:**Geometry tells the extractor **WHERE** the wires are; technology data tells it **HOW** they behave electrically.

Accurate Extraction = Reliable Timing

8

How does QRC / Quantus actually extract R and C ?

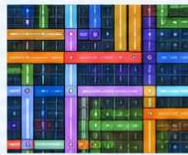
QRC (Cadence) or Quantus reads the routed layout and technology data, calculates the parasitic resistance and capacitance for every wire segment, and generates a distributed RC network, which is written to a SPEF file.

Layout
P.**1 Read Routed Geometry**

- QRC/Quantus reads the routed .DEF file along with LEF files.
- It gets the exact physical geometry of every wire, via and layer.
- Builds a 3D model of the interconnect structure.



Routed DEF



Routed Layout View (from DEF)

2 Read Technology RC Information

- Reads technology file (e.g., qrcTech / tech files).
- Gets metal layer properties:
 - sheet resistance
 - capacitance per unit length/area
 - via resistance
 - layer thickness, spacing, etc.

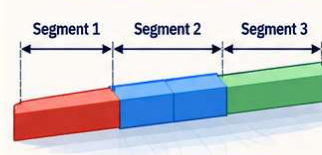


(Technology File)

Example (Tech Data)		
Layer	R _s (Ω/□)	C (fF/μm)
M1	0.38	0.035
M2	0.28	0.032
M3	0.25	0.030
...	...	...

3 Break Wires into Segments

- Each routed wire is divided into small segments based on:
 - changes in layer
 - width/spacing variations
 - corners / bends
 - vias
 - neighboring wires (for coupling)



Same wire, different segments (with layer/width changes)

4 Calculate Resistance

- For each segment, resistance is calculated using:

$$R = R_s \times \left(\frac{L}{W} \right)$$

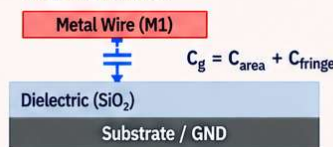
R_s : sheet resistance (Ω/□)
L : length of segment (μm)
W : width of segment (μm)

Example (M1 segment)

Length (L) = 0.22 μm
Width (W) = 0.02 μm
R_s (M1) = 0.38 Ω/□
Number of squares = 0.22 / 0.02 = 11
R = 0.38 × 11 = 4.18 Ω

5 Calculate Ground Capacitance (C_g)

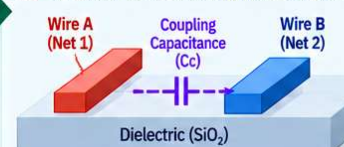
- Calculates capacitance from the wire to the reference ground (substrate / nearest ground plane).
- Includes area capacitance and fringe capacitance.



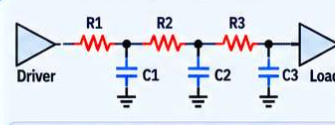
C_g depends on layer, width, thickness, dielectric constant and distance to ground.

6 Calculate Coupling Capacitance (C_c)

- Calculates capacitance between adjacent parallel wires.
- Depends on spacing, length of parallel run, and dielectric properties.
- Important for crosstalk and noise analysis.

**7 Build Distributed RC Network**

- Combines all segment resistances, ground capacitances and coupling capacitances.
- Creates a distributed RC network (multiple R and C values per net), instead of a single lumped value.

**8 Write SPEF**

- All computed R and C values are into a .SPEF file in IEEE 1481 format.
- One SPEF per RC corner (e.g., worst case).



Extracted SPEF (Parasitic R & C)

```
*D_NET net1 0.1
*CONN
*I U1/A 0 *I
*I U2/B I *I
*CAP
1 net1:1 0.
2 net1:2 0.
*RES
1 net1:1 ne
...
*END
```

**Key Takeaway:**

“QRC/Quantus converts the routed physical geometry into an accurate distributed R and C network using technology data, and writes it to a SPEF file for signoff timing analysis.”

Real Geo
Real Par
Accurate

9

How is Wire Resistance calculated from Physical Geometry ?

A metal wire has finite resistance. It depends on the wire length, width, metal layer properties and is calculated using sheet resistance and number of squares.

Longer
Higher
Wider
Lower**1 Formula**

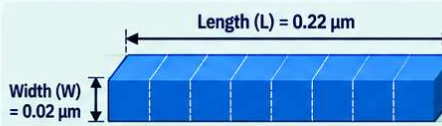
$$R = R_s \times \left(\frac{L}{W} \right)$$

R : Resistance of the wire segment (Ω)
R_s : Sheet resistance of the metal layer (Ω/□)
L : Length of the wire segment (μm)
W : Width of the wire segment (μm)
 $\frac{L}{W}$: Number of squares (dimensionless)

Resistance is proportional to length and inversely proportional to width.

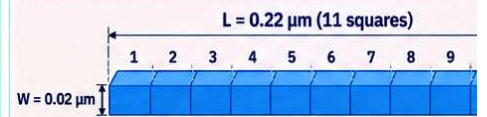
2 What is Number of Squares ?

- A “square” is a unit of resistance (length = width).
- Number of squares (N) = Length / Width
- It is independent of the actual size (e.g., 10 μm / 1 μm = 10 squares, 1 μm / 0.1 μm = 10 squares → same resistance).
- This makes calculation simple using sheet resistance.

**3 Example Calculation (M1 Wire)**

Length (L) = 0.22 μm
Width (W) = 0.02 μm
Sheet Resistance (R_s) = 0.38 Ω/□ (for M1, example)
Number of squares (N) = 0.22 / 0.02 = 11

Resistance (R) = 0.38 × 11 = 4.18 Ω

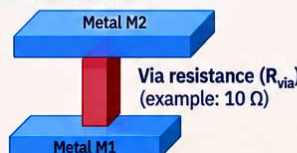
**4 Typical Sheet Resistance (R_s) Values**

Metal Layer	Sheet Resistance (Ω/□)
M1	0.38 (example)
M2	0.28
M3	0.25
M4	0.20
M5	0.18
M6	0.15
...	...

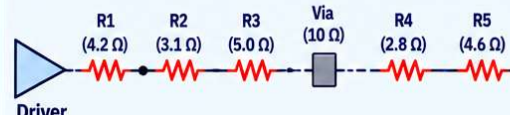
R_s depends on technology node, metal layer and process (see DEF/qrcTech file for exact values)

5 Via Resistance

- Vias also have resistance (not just metal wires).
- Each via has a fixed resistance value (provided in technology file).
- Total path resistance includes wire segment resistance + via resistances.

**6 From Segments to a Resistance Network**

- A real routed net is divided into many wire segments and vias.
- Resistance is calculated for each segment.
- All segment resistances and via resistances are connected in to form a distributed RC network (along with capacitances).



These resistances (with corresponding capacitances) form a distributed RC network.

and process (use QRC technology for exact values).

$$R_{\text{total}} = R_{\text{wire}(M1)} + R_{\text{via}} + R_{\text{wire}(M2)} + \dots$$

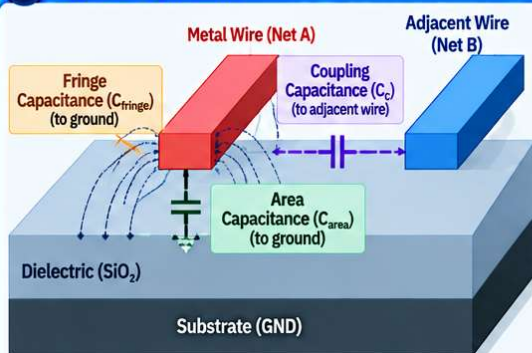
a distributed RC network, which is used for accurate timing a

**Key Takeaway:**

"Wire resistance = Sheet resistance × Number of squares.
Calculate it for each segment and include via resistance to get the **total resistance** of the routed net."

Accurate
Accurate
Reliable**10****How is Wire Capacitance Calculated ?**

Total wire capacitance is the combination of area capacitance, fringe capacitance (both to ground) and coupling capacitance (to neighboring wires).
It depends on the wire geometry (width, thickness, spacing, length), metal layer and the technology data (dielectric constants, RC models).

Cap
Del
Ge
Te**1 Types of Capacitance****2 Area Capacitance (C_{area})**

- Due to the parallel plate effect between the wire and the underlying ground plane.
- Proportional to wire area (length × width).
- Depends on dielectric constant (k) and oxide thickness (t_{ox}) of the layer.

$$C_{\text{area}} = C_{\text{pp}} \times L \times W$$

$$\text{where } C_{\text{pp}} = \frac{\epsilon}{t_{\text{ox}}} \text{ (per unit area)}$$

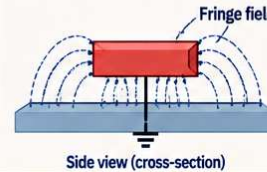
L : Length of wire (μm)
 W : Width of wire (μm)
 ϵ : Permittivity of dielectric (F/ μm)
 t_{ox} : Dielectric thickness (μm)

3 Fringe Capacitance (C_{fringe})

- Due to fringing electric fields from the sides and edges of the wire.
- Proportional to wire length.
- Becomes significant for higher metal layers and wider spacing.

$$C_{\text{fringe}} = C_f \times L$$

where C_f depends on layer, width, spacing and technology.

**4 Ground Capacitance**

- Sum of area capacitance and fringe capacitance to the reference ground (substrate or near).

$$C_g = C_{\text{area}} + C_{\text{fringe}}$$

This is the capacitance from the wire to the reference ground plane.

5 Coupling Capacitance

- Occurs between adjacent wires on the same or nearby layer.
- Depends on spacing, length, and relative position.
- Can cause crosstalk and noise.
- Increases when spacing is smaller.

Smaller spacing → Higher coupling capacitance

6 Influence of Geometry and Technology

Factor	Effect on Capacitance	Explanation
Wire length (L)	↑ Increases	Both area and fringe C are proportional to length.
Wire width (W)	↑ Increases	Larger width increases area C (fringe C also affected).
Metal layer	Varies	Higher metal layers have different dielectric environment (usually larger spacing → lower C).
Metal spacing	↓ Increases (coupling)	Smaller spacing increases coupling capacitance.
Adjacent wires	↑ Increases (coupling)	More nearby wires → higher total coupling C.
Dielectric constant (k)	↑ Increases	Higher k results in higher capacitance.
Dielectric thickness	↓ Decreases	Thicker dielectric reduces capacitance.
Layer environment	Varies	Proximity to ground shields, other metals, and structures affects total capacitance.

7 Example Calculation

Consider a metal wire on M1 layer with:
 Length (L) = 0.22 μm
 Width (W) = 0.02 μm
 Assume (example values):
 Area capacitance per unit area = 0.035 fF/ μm^2
 Fringe capacitance per unit length = 0.042 fF/ μm

1) Area capacitance:

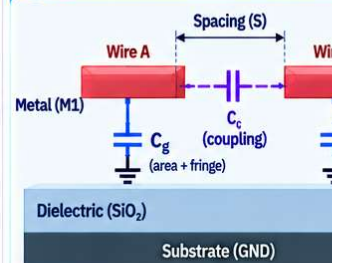
$$C_{\text{area}} = 0.22 \times 0.02 \times 0.035 = \mathbf{0.154 \text{ fF}}$$

2) Fringe capacitance:

$$C_{\text{fringe}} = 2 \times 0.22 \times 0.042 = \mathbf{0.0185 \text{ fF}}$$

3) Total ground capacitance:

$$C_g = 0.154 + 0.0185 = \mathbf{0.1725 \text{ fF}}$$

8 Simple Cross-section View**Key Takeaway:**

Wire capacitance depends on the **physical geometry** (length, width, spacing), the **layer environment** and the **technology dielectric properties**. Accurate RC extraction tools (QRC/Quantus) use this information to compute these capacitances for every wire segment.

Accurate Cap
Leads to Acc
and Better S**11****How do you perform post-route RC extraction in Innovus?**

In Innovus, post-route RC extraction is performed using the `extract_rc` command with appropriate options and QRC technology.

Accurate
Accurate
Reliable**1 Basic Command**

```
extract_rc \
-effort high \
-coupled true \
-qrc_config <qrcTech_path>
```

`extract_rc` : Command to run parasitic extraction
`-effort` : Sets extraction effort/accuracy
`-coupled` : Enables coupling capacitance extraction
`-qrc_config` : Specifies QRC technology configuration file

2 What is extract_rc ?

- Innovus command to perform parasitic extraction using Cadence QRC.
- Reads the routed design, technology data, and extraction settings.
- Extracts resistance and capacitance for wires, vias and devices.
- Generates an extracted RC database (in-memory) that can be used for STA.
- Can also write SPEF/RCF for signoff.

`extract_rc` runs QRC under the hood to compute the **real parasitics** of the routed design.

3 -effort Option

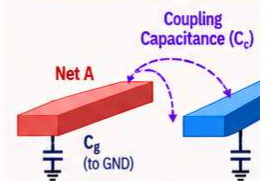
- Controls the accuracy vs runtime of extraction.
- Common values:

Value	Description
low	Faster runtime, lower accuracy
medium	Balanced runtime and accuracy (default)
high	Slower runtime, higher accuracy (recommended for signoff)

Use `-effort high` for final signoff extraction to get accurate parasitics.

4 -coupled Option

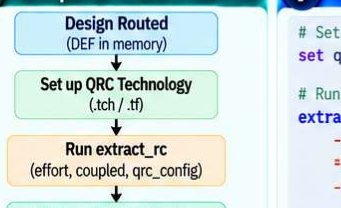
- `-coupled true` extracts both capacitance and coupling capacitance between neighboring nets.
- `-coupled false` extracts only capacitance (faster, but less accurate).



Use `-coupled true` for accurate post-route timing analysis

5 QRC Technology Configuration

- `-qrc_config` specifies the QRC technology configuration file (usually .tch or .tf).
- This file contains:
 - Layer stack information
 - RC models for each layer
 - Process/dielectric parameters
 - Extraction rules and options

**6 Complete Flow in Innovus****7 Example in Innovus TCL**

```
# Set QRC technology file path
set qrcTech "/path/to/tech/tech_qrc.tch"

# Run post-route RC extraction
extract_rc \
-effort high \
-coupled true \
-qrc_config $qrcTech
```

8 Sample Innovus Log (Console Output)

```
innovus> extract_rc -effort high -coupled true -qrc_config /path/to/tech/tech_qrc.tch
QRC-Info: Reading technology file ...
QRC-Info: Building RC models ...
QRC-Info: Extracting parasitics (effort: high, coupled: true) ...
QRC-Info: Extraction completed.
QRC-Info: RC data is available in the design database.
```


(QRC config file)

Use the correct QRC technology file provided by your foundry/PDK for your technology node (e.g., 28nm, 14nm, 7nm, etc.).

Extracted RC in Memory (used for STA)

Write SPEF (optional) (for signoff / external STA)

```
# (Optional) Write SPEF
write_spef -corner slow \
-compress \
-output top_postroute.spef
```

CPU Time : 00:12:34
Memory : 4.8 GB

innovus> ✓ Extraction Completed

Key Takeaway:

“Use **extract_rc** with **-effort high**, **-coupled true** and the correct QRC technology file to get **accurate post-route parasitics** for reliable timing analysis.”

✓ Real P
Accura
Signof

12

How do you generate a **SPEF** from the **extracted RC** database ?

After running parasitic extraction (**extract_rc**), the RC data is available in the Innovus database. Use the **rcOut** command to write the SPEF file for a specific RC corner.

1 Basic Command

```
rcOut \
-spef spef/mimo_rcworst.spef \
-rc_corner rcworst \
-noBackSlash
```

rcOut : Writes the extracted RC data to a file (SPEF/RCF).

-spef : Specifies the output SPEF file path.

-rc_corner : Selects the RC corner from the extracted database.

-noBackSlash : Option to control net name formatting (remove backslash).

2 What does rcOut do ?

- Reads the extracted RC database (in-memory) after **extract_rc**.
- Writes the parasitic data in SPEF format (IEEE 1481).
- Exports R and C for the specified RC corner.
- Can also write RCF or other formats (depending on options).
- Used for post-route STA in PrimeTime or other timing tools.

rcOut does not perform extraction.
It only writes the already extracted RC data to a file (SPEF).

3 SPEF Output Path (-spef)

- Specify the full or relative path for the output SPEF file.
- Make sure the directory exists.
- Use a meaningful file name (e.g., design name + corner).

Example:

```
-spef spef/mimo_rcworst.spef
```



Create the output directory before running **rcOut**.

4 RC Corner Selection (-rc_corner)

- Specifies which extracted RC write into the SPEF file.
- The corner must already be existing using **extract_rc**.
- Common corner names: **rcworst**, **rcbest** (as defined in QRC/Innovus).

Corner Name	Typical
rcworst	Worst case (max delay)
rc typical	Typical case
rcbest	Best case (min delay)

Run **rcOut** separately for each corner if you need multiple SPEF files.

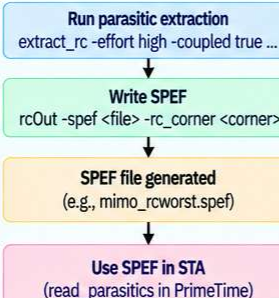
5 Naming Options (-noBackSlash)

- noBackSlash** removes backslashes (\) from hierarchical net names in the SPEF file.
- Useful for tools that do not support \ in net names.
- If not used, hierarchical names may appear as `\top/u1/net`.

Option	Effect
(default)	Keeps backslashes in net names
-noBackSlash	Writes net names without backslashes (e.g., <code>top/u1/net</code>)

Use **-noBackSlash** for better compatibility with most STA tools (e.g., PrimeTime).

6 Complete Flow in Innovus



7 Example Innovus Session

```
innovus> extract_rc -effort high -coupled true \
-qrc_config /path/to/tech_qrc.tch
QRC-Info: Extracting parasitics ...
QRC-Info: Extraction completed.

innovus> rcOut -spef spef/mimo_rcworst.spef \
-rc_corner rcworst -noBackSlash
QRC-Info: Writing SPEF file ...
QRC-Info: SPEF file 'spef/mimo_rcworst.spef' generated successfully.

innovus> ✓ SPEF generated successfully!
```

8 SPEF File (Partial View)

```
*SPEF "IEEE 1481-1998"
*DESIGN "mimo"
*DATE "Sep 15 2026"
*VENDOR "Cadence"
*PROGRAM "Innovus QRC"
*VERSION "23.1"

*D_NET top/u1/net1 0.1234
*CONN
*I U1/A
*I U2/Z
*CAP
1 0.0234
2 0.0456
*RES
1 2.3456
2 1.9876
*END
```

Key Takeaway:

“Use **rcOut** with the correct output path, RC corner and naming options to generate a clean SPEF file from the extracted RC database for accurate post-route timing analysis.”

✓ Extracted Para
Reliable STA =

13

How do you run **standalone QRC** extraction ?

Standalone QRC is a Cadence tool used to extract parasitic R and C from a routed design using technology files and layout data, without launching Innovus.

1 Command to Run Standalone QRC

```
qrc \
-cmd qrc_settings.tcl \
-log qrc_extraction.log
```

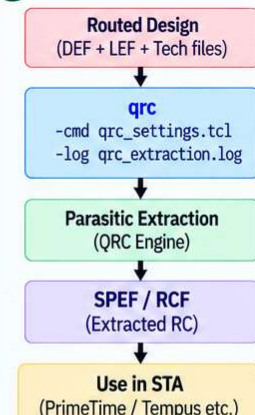
qrc : Invokes the standalone QRC tool

-cmd : Specifies the TCL command file with extraction settings

-log : Specifies the log file to capture the extraction messages

Make sure **qrc_settings.tcl** contains all the required inputs and extraction options.

2 Overall Flow



3 Key Settings in qrc_settings.tcl

```
# -----
# QRC Extraction Settings
# -----

# Read input files
set tech_file "/tech/qrcTechfile"
set def_file "/input/design.def"
set lef_file "/lef/tech.lef /lef/stdcell.lef"

# Set extraction options
set_extraction_options \
-qrc_tech_file $tech_file \
-def_file $def_file \
-lef_file $lef_file \
-net_name_space "" \
-coupling_cap true \
-effortLevel high \
-output_file "/output/design.spef" \
-format SPEF
```

4 Important Settings Explained

Setting	Description
set_extraction_options	Main command to set extraction parameters
-qrc_tech_file	QRC technology file (.tch / .tcf)
-def_file	Routed design DEF file
-lef_file	LEF files (tech LEF, standard macro LEF, etc.)
-net_name_space	Hierarchy separator (e.g., "/" or ".")
-coupling_cap	true = extract coupling capacitance, false = ground capacitance
-effortLevel	Extraction accuracy (low medium high)
-output_file	Output file path (e.g., design.spef)
-format	Output format (SPEF, RCF, etc.)

Use **-coupling_cap true** and **-effortLevel high** for accurate signoff extraction.

5 Input Files for QRC

	design.def	Contains final routed geometry (wires, vias, etc.)
--	------------	--

6 Example Run (Terminal)

```
$ qrc -cmd qrc_settings.tcl -log qrc_extraction.log
QRC> Reading command file: qrc_settings.tcl
```

7 Output Files

Extracted parasitic R and C in SPEF format

8 Tips for Accurate Extraction

- Use the correct QRC technology file (for your process node).

Technology LEF	tech.lef	Metal layers, via definitions, layer properties
Standard Cell LEF	stdcell.lef	Cell geometries and pins
Macro LEF (SRAM/IP)	macro.lef	Macro geometries (if any)
QRC Tech File	qrcTech.tch (or .tch)	Technology RC information for extraction (layer, via, cap models)

QRC> Reading technology file ...
 QRC> Reading LEF/DEF files ...
 QRC> Setting up extraction ...
 QRC> Extracting RC (effort: high, coupling: true) ...
 QRC> Writing SPEF: ./output/design.spef
 QRC> Extraction completed successfully.

QRC extraction completed successfully!

design.spef (used for STA)

design.rcf Alternative output format (Cadence RCF)

design.log Log file with extraction details, warnings and runtime info

Accurate extraction = Accurate Reliable signoff!

Key Takeaway:

“Run standalone QRC using the **qrc** command with a proper **TCL settings file**, correct input files, and appropriate extraction options to generate an **accurate SPEF** for post-route STA.”

Correct In
Correct S
Accurate

What is the difference between QRC/Quantus and StarRC?

Both are signoff parasitic extraction solutions from different EDA vendors. They have the same fundamental purpose (extracting R and C from a physical layout), but they use different technology/configuration ecosystems and integrate with their respective tool flows. They can both generate industry-standard formats such as SPEF.

1 Cadence Quantus / QRC Flow

Routed Design + Technology Data

- Routed DEF (with full routing)
- LEF files (tech + stdcell + macros)
- QRC Technology Files (.tch / .tf)
- Extraction settings (TCL)

→ **Quantus / QRC** Parasitic Extraction → **Extracted RC** (SPEF / RCF Parasitics) → **Used in Cadence Flow** (Innovus (Implementation), Tempus (STA / Signoff))

Cadence describes Quantus as its signoff parasitic extraction solution integrated with Innovus and Tempus.

2 Synopsys StarRC Flow

Layout + Technology Data

- Routed DEF (with full routing)
- LEF files (tech + stdcell + macros)
- StarRC Technology Files (.tlup, .ict, etc.)
- Extraction settings (TCL)

→ **StarRC** Parasitic Extraction → **Extracted RC** (SPEF / RCF Parasitics) → **Used in Synopsys Flow** (ICC2 (Implementation), PrimeTime (STA / Signoff))

Synopsys describes StarRC as its signoff parasitic extraction solution supporting industry-standard formats and integrating with its implementation, timing, SI, power, physical verification and simulation flows.

3 Same Fundamentals

- Both tools extract parasitic (R) and capacitance (C) from layout (routed design).
- Used for timing analysis, signal integrity (SI), power analysis and signal integrity (SI).
- Both consider real layout geometry, layer stack and process technology information.
- The extraction algorithms may differ, but the objective is to generate accurate parasitics for signoff.

4 Key Differences (High Level)

- Different EDA vendors: Cadence (Quantus/QRC) vs Synopsys (StarRC).
- Different technology/configuration files (e.g., QRC tech files vs StarRC TLUP/ICT files).
- Different tool ecosystems and integration (Quantus with Innovus/Tempus, StarRC with ICC2/PrimeTime and other Synopsys tools).
- Different extraction commands and settings.
- Both support industry-standard formats such as SPEF, enabling interoperability (with proper naming, corners, units, etc.).

5 Side-by-Side Comparison

Aspect	Cadence Quantus / QRC	Synopsys StarRC
Vendor	Cadence	Synopsys
Extractor	Quantus / QRC	StarRC
Typical implementation integration	Innovus (place and route)	ICC2 (place and route)
Typical STA integration	Tempus	PrimeTime
Technology / configuration ecosystem	QRC technology files (.tch / .tf) and Quantus settings	StarRC technology files (.tlup, .ict, etc.) and StarRC settings
Standard output example	SPEF, RCF (IEEE 1481)	SPEF, RCF (IEEE 1481)
Core purpose	Signoff parasitic extraction (for timing, SI, power, etc.)	Signoff parasitic extraction (for timing, SI, power, etc.)

6 Integration with STA Tools

Cadence Flow: Quantus / QRC (Extraction) → SPEF / RCF → STA Tools

Synopsys Flow: StarRC (Extraction) → SPEF / RCF → STA Tools

Both tools can integrate with broader industry-standard formats and support standard formats such as SPEF for interoperability.

7 What is NOT the Difference?

- It is NOT that one extracts parasitics and the other does not.
- It is NOT that one is only for post-route and the other is for pre-route.
- It is NOT that they produce completely different output formats.
- It is NOT that one is inherently more accurate in all cases.

The fundamental extraction objective is the same. The main differences are the ecosystem, technology files, configuration, interfaces and integration with EDA flows.

8 Standard Format Interoperability (e.g., SPEF)

Both Quantus and StarRC can generate industry-standard exchange formats such as SPEF.

SPEF can be transferred between compatible tools (e.g., used in Tempus, PrimeTime or other tools), subject to:

- Consistent net naming
- Matching corners and units
- Flow and tool compatibility

9 Key Takeaway

Same extraction, Different EDA ecosystem, Standard format: enable interoperability.

Takeaway:

“QRC/Quantus and StarRC both extract parasitics for signoff. The difference lies in their **vendor ecosystem, technology/configuration files, and tool integration**, while **standard formats such as SPEF** enable interoperability across EDA tools.”

Accurate Parasitics
Enable Accurate
Better Chip

15 What is the complete internal structure of a SPEF file?

A SPEF (Standard Parasitic Exchange Format) file contains parasitic information (R and C) for a design in a defined hierarchical structure.

1 Major Architecture of a SPEF File

- HEADER** (File information, units, options)
- NAME_MAP** (Compressed name mapping)
- PORTS** (Top-level ports)
- D_NET** (Parasitic data for each net)
 - CONN (Node connections)
 - CAP (Capacitance values)
 - RES (Resistance values)
 - END (End of this net)
- Next D_NET** (Repeat for all nets)

2 Description of Each Section

- HEADER**
 - Contains file information, version, date, design name, units, and options.
 - Defines the format and settings used for extraction.
- NAME_MAP**
 - Maps compressed names (used in the file) to actual hierarchical names.
 - Reduces file size by using short names.
- PORTS**
 - Lists all primary input/output/inout ports of the design.
 - Includes port direction and optional capacitance.
- D_NET (for each net)**
 - Contains the parasitic information for a single net.
 - Consists of the following subsections:
 - CONN** Lists all the connected nodes (pins, internal nodes, etc.).
 - CAP** Contains capacitance values (ground and coupling).
 - RES** Contains resistance values between nodes.
 - END** Marks the end of the current net.

3 Example SPEF File (Simplified)

```

* S P E F "IEEE 1481-1998"
* DESIGN "mimo_top"
* DATE "Sep 15 2026"
* VENDOR "Cadence QRC"
* VERSION "23.1"
* DIVIDER /
* DELIMITER :
* BUS_DELIMITER []
* T_UNIT 1.0 NS
* C_UNIT 1.0 FF
* R_UNIT 1.0 OHM
* L_UNIT 1.0 NH
* NAME_MAP "YES"

* NAME_MAP
*1 U1/A
*2 U1/B
*3 U2/Z
*END

* PORTS
*1 in1
*0 out1
*0 out2
*END

* D_NET *1 (in1)
* CONN
*1 I
*2 U1/A
*3 U2/Z
* CAP
*1 0.0234
*2 0.0456
* RES
*1 2.3456
*2 1.9876
* END
            
```


6

END
(End of SPEF file)

5

NEXT D_NET
• The D_NET section repeats for every net in the design.

6

END
• Marks the end of the SPEF file.

*D_NET *2 (out1)
...
*END

}
Next (Repeat)
End of

Key Takeaway:

"A **SPEF** file has a well-defined **hierarchical structure**: **HEADER** → **NAME_MAP** → **PORTS** → **D_NET** (with **CONN**, **CAP**, **RES**, **END**) repeated for each net → **END**."

Standard
Tool Inter
Used for S

16

What information is present in the **SPEF** header ?

The SPEF header contains file identity, tool/version, design details, naming rules, units, and other information for compatibility and debugging.

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SPEF Header Fields and Their Meaning

Field	What it Contains	Category
*SPEF	SPEF format version (e.g., "IEEE 1481-1998")	File Identity
*DESIGN	Top-level design name	Design
*DATE	Date and time when the file was generated	File Identity
*VENDOR	EDA vendor (e.g., Cadence, Synopsys)	Tool / Version
*PROGRAM	Tool used to generate the file (e.g., QRC, Quantus, StarRC)	Tool / Version
*VERSION	Version of the tool	Tool / Version
*DESIGN_FLOW	Flow stage (e.g., post-route, signoff)	Compatibility / Debug Info
*DIVIDER	Hierarchical name divider (e.g., "/")	Naming Rules
*DELIMITER	Character used to separate tokens (e.g., space)	Naming Rules
*BUS_DELIMITER	Bus naming delimiters (e.g., "[]")	Naming Rules
*T_UNIT	Time unit (e.g., 1.0 NS)	Units
*C_UNIT	Capacitance unit (e.g., 1.0 FF)	Units
*R_UNIT	Resistance unit (e.g., 1.0 OHM)	Units
*L_UNIT	Inductance unit (e.g., 1.0 NH)	Units

2

Example SPEF Header (Realistic)

```
*SPEF      "IEEE 1481-1998"  
*DESIGN    "mimo_top"  
*DATE      "Sep 15 2026 14:23:10"  
*VENDOR    "Cadence"  
*PROGRAM   "Quantus QRC"  
*VERSION   "23.1"  
*DESIGN_FLOW "post_route"  
*DIVIDER   "/"  
*DELIMITER "  
*BUS_DELIMITER "[ ]"  
*T_UNIT    1.0 NS  
*C_UNIT    1.0 FF  
*R_UNIT    1.0 OHM  
*L_UNIT    1.0 NH
```

This header tells us:

- It is a SPEF file (IEEE 1481-1998)
- For design "mimo_top"
- Generated by Cadence QRC (version 23.1)
- At post-route stage
- Uses "/" as hierarchy divider and "[]" for buses
- Units: time (NS), capacitance (FF), resistance (OHM), inductance (NH)

The header is read by STA tools (e.g., PrimeTime) to correctly interpret units and ensure compatibility.

3

Information Categories (Summary)

File Identity
Format, date, file information

Design
Top-level design name

Tool / Version
EDA vendor, tool name and version

Naming Rules
Hierarchy divider, delimiters, bus notation

Units
Time, capacitance, resistance, inductance units

Compatibility / Debug Info
Design flow and other useful information

4

Why the Header is Important ?

- Helps STA tools parse the file correctly.
- Provides design and tool information for debug and traceability.
- Ensures consistent interpretation of units across tools.
- Useful for verifying that the SPEF matches the design and flow stage.
- Helps identify any mismatches or version conflicts.

Key Takeaway:

"The **SPEF** header provides **essential information** about the file, design, tool, naming rules, units, and flow stage, ensuring correct interpretation and compatibility across EDA tools."

Correct H
Correct In
Reliable S

17

What is ***NAME_MAP** and why is it required ?

SPEF files use integer aliases to represent long hierarchical names. This mapping is defined in the *NAME_MAP section, which reduces file size and makes the file easier to parse.

Save
Sw
Better

1

What is *NAME_MAP ?

- *NAME_MAP is a section in the SPEF file that maps integer IDs (aliases) to actual hierarchical net or instance names.
- Instead of using long hierarchical names everywhere in the file, SPEF uses short integer aliases (e.g., *1, *2, *3, ...).
- This mapping allows the rest of the file (e.g., in D_NET, CONN, etc.) to reference names using these integer IDs.
- It is basically a look-up table for names.

2

Why is it required ?

- Long hierarchical names**
Modern SoCs have very long hierarchical names (hundreds of characters).
- Compression**
Using integer aliases significantly reduces file size.
- Faster parsing**
Easier and faster for tools to read and process the file.
- Consistent referencing**
Ensures the same names are used consistently throughout the file.
- Industry standard**
Defined in IEEE 1481 SPEF format.

3

Example: *NAME_MAP Section

```
*NAME_MAP  
*1      mimo_top:VDD  
*2      mimo_top:VSS  
*3      clk_gen/clk  
*4      u_dma/wr_ptr_reg[3]  
*5      fft_core/butterfly_reg_0  
*6      u_ctrl_fsm/state_reg[0]  
*7      top/u_mem/mem_array[127]  
*8      u_axi/awaddr[15]  
*9      u_data_path/u_adder/sum  
*10     u_out_reg/Q  
*END
```

Each integer (e.g., *1, *2, *3, ...) is an alias for a long hierarchical name.

4

How it is used in the rest of the file

```
*D_NET *9  
*CONN *0 *5 I  
*1 *10 I  
*CAP *1 0.0234  
*2 0.0456  
*RES *1 2.3456  
*2 1.9876  
*END
```

Here, ***D_NET *9** means the net ID ***9**, which corresponds to the alias **u_data_path/u_adder/sum** (lookup in *NAME_MAP):
***9 → u_data_path/u_adder/sum**

5

Without *NAME_MAP (Not Recommended)

If integer aliases were not used, the full hierarchical names would appear everywhere in the file:
*D_NET u_data_path/u_adder/sum
*CONN

6

With *NAME_MAP (Recommended)

Using integer aliases keeps the file compact and clean:
*D_NET *9
*CONN

7

Mapping Illustration

Long Hierarchical Names (in design)

*NAME_MAP (Integer Aliases)

Use and of

mimo_top:VDD → *1 → *D

https://onedrive.live.com/personal/938043310494b6ef/_layouts/15/Doc.aspx?sourcedoc={0494b6ef-4331-2080-8093-7a0000000000}&action=edit&w... 8/17


```

*CONN
u_data_path/u_reg1/Q I
u_data_path/u_reg2/Q I
...
*END

```

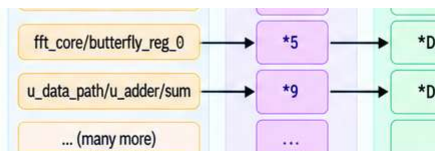
This increases file size, makes the file hard to read, and slows down parsing.

```

*CONN
*5 I
*10 I
...
*END

```

Same information, but much smaller and easier to process!



Key Takeaway:

"*NAME_MAP maps **integer aliases** to long hierarchical names, **reducing file size** and making the SPEF file **easier and faster to process**."



- Smaller File
- Faster Parsing
- Standard & T

18

What is the *PORTS section ?

The *PORTS section in a SPEF file lists all the top-level ports of the design, their direction (input/output/bidirectional) and the associated port capacitance.

Shows how connects v

1 What is the *PORTS section ?

- The *PORTS section lists all the primary (top-level) input/output/inout ports of the design.
- Each port is specified with its name, direction and parasitic capacitance connected to that port.
- It represents the external interface of the design in the SPEF file.
- This information is used by STA tools to accurately model the port load and boundary conditions.

2 Syntax and Example

***PORTS**
*<index> <direction> *C <cap_value>

Example:

```

*PORTS
*1 clk I *C 0.0234
*2 rst_n I *C 0.0187
*3 data_in I *C 0.156
*4 data_out O *C 0.0945
*5 io_pad B *C 0.210
*END

```

3 Meaning of Each Field

Field	Description
*<index>	Integer ID from *NAME_MAP that represents the port name (e.g., *3 → data_in)
<direction>	Port direction I = Input O = Output B = Bidirectional (Inout)
*C	Indicates that the following value is a capacitance
<cap_value>	Total capacitance (in SPEF C_UNIT, (e.g., 1.0 FF) seen at the port)

4 Port Direction Type

Direction	Meaning	Typical
I	Input	Primary input (e.g., clk, rst_n)
O	Output	Primary output (e.g., data_out)
B	Bidirectional	Inout port (e.g., shared buses)

Direction is defined with respect to (i.e., from the design's point of view)

5 What does Port Capacitance Represent ?

- Port capacitance is the total parasitic capacitance seen at the top-level port.
- It includes the capacitance of the port net, routing, pads and any connected structures inside the design.
- It models the external load at the boundary of the design.
- This capacitance is extracted from the layout during RC extraction and written into the SPEF file.

6 Example with Name Mapping

***NAME_MAP (partial)**

```

*1 clk
*2 rst_n
*3 data_in
*4 data_out
*5 io_pad
*END

```

***PORTS section**

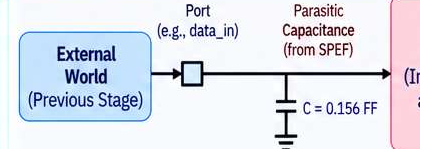
```

*PORTS
*1 I *C 0.0234
*2 I *C 0.0187
*3 I *C 0.156
*4 O *C 0.0945
*5 B *C 0.210
*END

```

Here, *3 I *C 0.156 means the port "data_in" is an input port with 0.156 (in C_UNIT, e.g., FF) capacitance.

7 How STA Uses Port Capacitance ?



- ✓ Used by STA to calculate accurate input delay and slew at
- ✓ Helps in timing analysis, especially for top-level paths.
- ✓ Ensures correct modeling of external load at the design boundary
- ✓ Important for signoff timing and for correlation with back-annotated



Key Takeaway:

"The *PORTS section defines the **top-level design ports**, their direction (I/O/B) and the associated **parasitic capacitance**, which is used by STA tools to model the external load and perform accurate **timing analysis**."



- Accurate Timing
- Correct Boundary
- Essential for Signoff

19

What is *D_NET ? — The Heart of SPEF

*D_NET represents a distributed net with its parasitic RC network, and it is the most important section in a SPEF file.

A real just on it's a

1 What is *D_NET ?

- *D_NET defines a **distributed net** and its parasitic RC network.
- It contains the **net name**, total capacitance, and detailed parasitic information (connections, caps and resistors).
- It is the **heart** of the SPEF file because it models the actual interconnect parasitics of a net extracted from layout.
- Each net in the design has **one** *D_NET section.

2 Syntax and Example

Syntax:

*D_NET <net_name> <total_capacitance>

Example:

```
*D_NET *9 0.3847
```

- *9 → net name (integer alias from *NAME_MAP)
- 0.3847 → total capacitance of the net (in C_UNIT, e.g., pF or fF)

3 Internal Structure of a *D_NET Section

```

*D_NET *9 0.3847      ← Net name and total capacitance
*CONN
*1 I                  ← List of connected nodes (pins, internal nodes, etc.)
*2 O
...
*CAP
*1 0.0345             ← Capacitance at each node (ground and to other nodes)
*2 0.0567
...
*RES
*1 *2 0.1234          ← Resistance between nodes
*2 *3 0.0987
...
*END                  ← End of this net section

```

4 Why Not Just One Lumped R and C ?

- Real interconnects are distributed.
- Wires have resistance and capacitance along their length, not just at one point.

5 Relationship with *CONN, *CAP and *RES

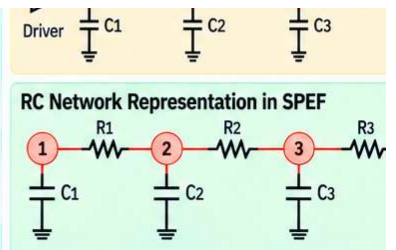
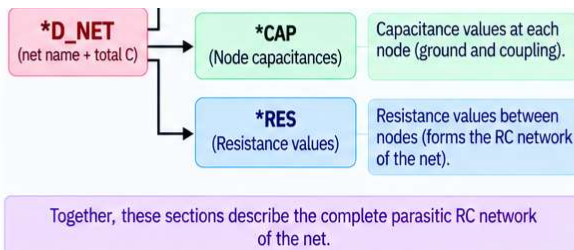
***CONN** (Connected nodes)
Lists all the nodes connected to this net (pins, internal nodes, etc.).

6 Visual Representation

Physical Net (Interconnect)



- Signal delay, slew and crosstalk depend on how R and C are distributed.
- A single lumped R and C cannot capture:
 - Elmore delay
 - Signal attenuation
 - Crosstalk between adjacent nets
 - Accurate timing for long wires and complex routing
- Therefore, SPEF stores a detailed RC network instead of one lumped value.



Key Takeaway:

"*D_NET represents a distributed net with its total capacitance and detailed RC network, using *CONN, *CAP and *RES sections, because real interconnects cannot be modeled accurately with a single lumped R and C."



- Accurate RC Modeling
- Better Timing Analysis
- Captures Crosstalk and

20

What is the *CONN section inside *D_NET?

The *CONN section inside *D_NET describes the connectivity of the net — which pins, ports, and instances are connected, their direction and cell information.

Conn
Def
Net's

1 What is *CONN?

- The *CONN section lists all the nodes connected to the net.
- It identifies whether the node is an instance pin (*I) or a top-level port (*P).
- It specifies the direction of the pin (Driver or Receiver/Load).
- It also provides the cell type and optional coordinates/offset of the node.
- This helps STA tools understand the connectivity and build the RC network correctly.

2 Syntax and Example

Syntax:

*I|*P <node_name> <direction> <coord> <cell_type>

Example:

*I *5:ZN 0 *L 0.00 *D SDFFRNQ_X1_LVT

Meaning:

- *I → Instance pin
- *5:ZN → Pin name (instance ID *5, pin ZN)
- 0 → Direction (Output, i.e., driver)
- *L → Location information (not used, default)
- 0.00 → Coordinate/offset (optional)
- *D → Indicates cell type follows
- SDFFRNQ_X1_LVT → Cell type of the instance

3 Field Description

Field	Description	Possible Values / Example
*I	Indicates an internal instance pin	*I (instance)
*P	Indicates a top-level port	*P (port)
<node_name>	Pin or port name (instance:pin or port)	*5:ZN, clk, rst_n
<direction>	Pin direction w.r.t. the net	I = Input O = Output (Driver) B = Bidirectional
<coord>	Coordinate/offset of the node	*L 0.00 (or actual coordinates)
*D	Indicates that cell type follows	*D
<cell_type>	Cell type of the instance (if *I)	SDFFRNQ_X1_LVT

4 Direction Meaning

Direction	Meaning
I	Input to the (Receiver/Load)
O	Output from (Driver)
B	Bidirectional (e.g., inout pins)

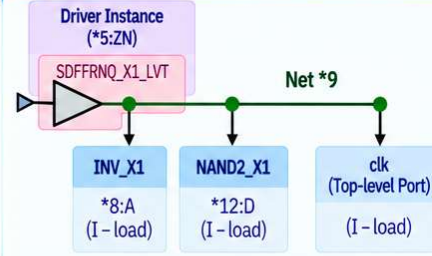
✓ Driver: Output pin (O) the signal to the net.
 ✓ Receiver/Load: Input consumes the signal.
 ✓ Bidirectional (B): Can input and output (e.g., shared buses).

5 Example *CONN Section in Context

```
*D_NET *9 0.3847
*CONN
*I *5:ZN 0 *L 0.00 *D SDFFRNQ_X1_LVT
*I *8:A I *L 0.00 *D INV_X1
*I *12:D I *L 0.00 *D NAND2_X1
*P clk I *L 0.00
...
*END
```

Here, net *9 is driven by instance *5 (pin ZN) and connects to other instance pins and a top-level port (clk).

6 Visual Representation



7 Key Takeaways

- *CONN defines the connectivity of the net.
- Identifies drivers, loads, internal pins and top-level ports.
- Uses *I for instance pins and *P for ports.
- Specifies pin direction (I/O/B).
- Includes cell type information for instance pins (using *D).
- Coordinates/offset (*L, value) are optional.
- Helps STA tools build an accurate RC net and perform correct timing analysis.



Key Takeaway:

"The *CONN section inside *D_NET lists all connected nodes (instance pins and ports), identifies the driver and loads, and provides direction and cell type information to help STA tools build the correct RC network of the net."



- Accurate Connectivity
- Identifies Driver & Loads
- Helps STA & Timing

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How do you interpret a real *CONN line?

A *CONN line contains multiple fields that describe a connected node (instance pin or port), its direction, location and cell information.

Each field
specifically
informs
the connectivity

Example *CONN line:

*I *5:ZN 0 *L 0.00 *D SDFFRNQ_X1_LVT

1 Node Type	2 Instance ID	3 Pin Name	4 Direction	5 Coordinate / Offset	6 Offset Value	7 Cell Type Indicator	8 Cell Name
*I	*5	:ZN	0	*L	0.00	*D	SDFFRNQ_X1_LVT
Indicates an instance pin (as opposed to a top-level port).	Integer alias for the instance name (from *NAME_MAP section).	Pin name of the instance. (':' separates instance name and pin name).	Direction of the pin w.r.t. the net. I = Input O = Output B = Bidirectional	Location type and offset. *L → use location information (default) The actual coordinate	Coordinate / offset value of the pin (typically in database units, e.g., microns). 0.00 → pin location	Indicates that the cell type (master name) follows.	Library cell name of the instance

*I = instance pin
*P = top-level port

→ refers to a specific instance (e.g., u_ff1)

ZN → output pin of the cell

0 indicates this is a driver (output).

The default coordinate value is given in the next field.

offset (optional, can be non-zero)

Used only for instance pins (*I).

→ D trip-t (from stan library)

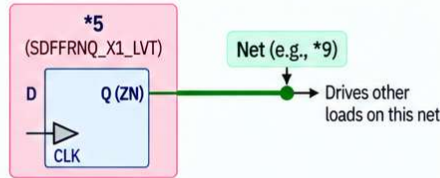
9 Putting it all together

*I *5:ZN 0 *L 0.00 *D SDFFRNQ_X1_LVT

This line means:

The pin ZN of instance *5 (a SDFFRNQ_X1_LVT cell) is connected to the net, it is an output pin (driver), and its location offset is 0.00.

10 Visual Representation



11 Key Takeaways

- ✓ *CONN describes connectivity at the pin
- ✓ *I and *P distinguish instance pins and top-level ports
- ✓ Direction (I/O/B) shows whether the pin is an input, output, or bidirectional
- ✓ Cell type (*D) and master name help identify the cell
- ✓ Coordinates are optional but can be used for placement



Key Message:

"A *CONN line provides a complete description of a connected node — what it is, where it comes from, its direction, location and cell type — helping STA tools accurately model the net's parasitics and connectivity."



- Clear Connectivity
- Identify Drivers
- Useful for STA analysis

22

What is the *CAP section ?

The *CAP section lists the parasitic capacitances at each node of the net, including ground capacitance and coupling capacitance.

Parasitics make the

1 What is *CAP ?

- The *CAP section lists the extracted parasitic capacitances at each node of the net.
- It includes both **ground capacitance** and **coupling capacitance**.
- Capacitances are reported **per node** (integer ID) in the RC network.
- Each node can have multiple capacitors connected to ground or to other nodes (coupling).
- The sum of all capacitances in this section equals the total capacitance reported in the *D_NET line.

2 Syntax and Example

Syntax:

*CAP

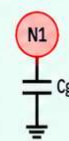
* <node1> <node2> <cap_value>

Example:

```
*CAP
*1 0.0234
*2 0.0456
*3 *4 0.0123
*3 0.0312
*4 *5 0.0087
...
```

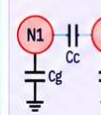
3 Types of Capacitance

1 Ground Capacitance (Node to Ground)



- Connects a node to ground (0).
- Represents capacitance to substrate, well, or ground reference.
- Caused by wire-to-ground, via-to-substrate, device diffusion, etc.
- Example line: *1 0.0234

2 Coupling Capacitance (Node to Node)



- Connects two different nodes.
- Represents capacitance between adjacent wires (crosstalk).
- Can affect timing due to coupling effects (victim/aggressor).
- Example line: *3 *4 0.0123

4 Field Description

Field	Description
*	Start of a capacitance entry
<node1>	First node ID (integer from RC network)
<node2>	Second node ID (0 = ground, or another node for coupling)
<cap_value>	Capacitance value (in C_UNIT, e.g., fF)

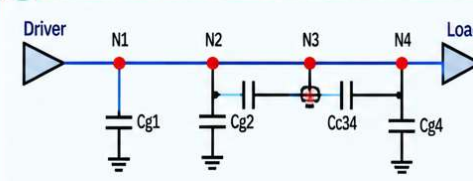
Units Capacitance values are in the unit in the SPEF header (*C_UNIT), typically femtofarads.

5 Example with Net Context

```
*D_NET *9 0.3847 ← Total capacitance of this net
*CONN
...
*CAP
*1 0.1000 (to ground)
*2 0.1200 (to ground)
*3 *4 0.0800 (coupling)
*4 0.0847 (to ground)
...
```

The sum of all capacitances in the *CAP section (0.1000 + 0.1200 + 0.0800 + 0.0847 = 0.3847) matches the total capacitance in *D_NET.

6 Visual Representation



Each node has ground capacitance, and adjacent nodes have coupling capacitance due to proximity.

7 Key Takeaways

- ✓ *CAP lists node-based parasitic capacitance
- ✓ Includes both ground and coupling capacitance
- ✓ Multiple capacitors exist because real interconnects have distributed parasitics (to ground and between nodes)
- ✓ Coupling capacitance models crosstalk effects
- ✓ The total capacitance in *D_NET equals the sum of capacitances in the *CAP section
- ✓ Used by STA tools to build the RC network for accurate timing and crosstalk analysis



Key Message:

"The *CAP section provides the detailed capacitance information at each node — both to ground and between nodes — and the sum of these capacitances equals the total capacitance reported in *D_NET."



- Models Real Parasitics
- Captures Crosstalk
- Essential for Accurate STA

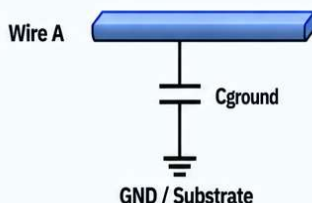
23

What is the difference between Ground Capacitance and Coupling Capacitance ?

Ground capacitance is the capacitance from a wire to the reference (ground/substrate), while coupling capacitance is the capacitance between two adjacent wires.

1 Ground Capacitance (Cground)

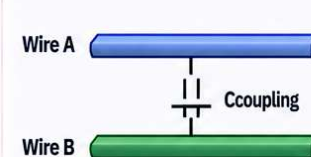
Capacitance between a wire and the reference (ground/substrate).



- Formed between the wire and the substrate/ground/well.
- Represents how much charge the wire stores with respect to ground.
- Always present for every wire.
- Contributes to the total load seen by the driver.
- Extracted as capacitance to ground in SPEF (e.g., *3 0.0234).

2 Coupling Capacitance (Ccoupling)

Capacitance between two adjacent wires (wire-to-wire).



- Formed between two nearby wires (due to electric fields).
- Represents the electric field interaction between the wires.
- Can cause crosstalk (noise) when one wire switches.
- Can be positive or negative impact depending on switching.
- Extracted as coupling capacitance in SPEF (e.g., *3 *4 0.0123).

Cground = capacitance from the wire to the reference (ground).

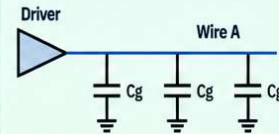
Ccoupling = capacitance between two wires.

3 Key Differences

Aspect	Ground Capacitance	Coupling Capacitance
What it is	Wire to ground/substrate	Wire to another wire
Representation	Cground	Ccoupling
Visual	Wire to ground	Between two wires
Effect	Adds to load capacitance	Causes crosstalk (noise)
Always present?	Yes (for every wire)	Only if nearby wires exist
Main impact	Affects delay (load)	Affects signal integrity (SI)
In SPEF	Node to ground (e.g., *3 0.0234)	Node to node (e.g., *3 *4 0.0123)
Nature	Always positive	Can be positive or negative (effective)

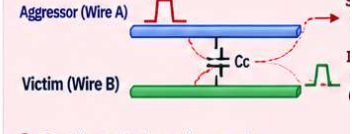
4 Impact on Signals

Ground Capacitance → Delay



- Increases the total load on the driver.
- Causes larger RC delay.
- Always considered in STA.

Coupling Capacitance → Crosstalk (SI)



- Can inject noise into adjacent wires.
- Causes crosstalk: glitch, delay shift, or even functional failure.
- Critical for signal integrity (SI) analysis (e.g., noise margin).
- Not simply a part of load capacitance.



Key Takeaway:

"Ground capacitance is a **wire-to-ground** capacitance that adds to the load and affects delay, while **coupling capacitance** is a **wire-to-wire** capacitance that causes crosstalk and impacts signal integrity — they serve different purposes and should not be confused."



- Ground Capacitance → Delay
- Coupling Capacitance → Crosstalk
- Both are Important in Real World

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How do you interpret a coupling-capacitance entry?

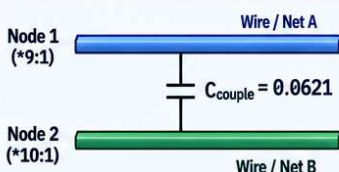
A coupling-capacitance entry lists the two nodes that are capacitively coupled and the value of the coupling capacitance between them.

Example Coupling-Capacitance Entry



1 ID	2 Node 1	3 Node 2	4 Coupling Capacitance
5	*9:1	*10:1	0.0621
- Unique identifier for this coupling entry. - Used by the tool for reference.	- First node of the coupled capacitor. - Node name (from *NAME_MAP) with optional pin/index (:1). - Represents a specific net/segment.	- Second node of the coupled capacitor. - Node name (from *NAME_MAP) with optional pin/index (:1). - Represents another specific net/segment.	- Coupling capacitance value between Node 1 and Node 2. - Given in the units specified in the SPEF header (typically fF). - This is a mutual capacitance, not to ground.

Visual Representation



Why can the coupled relationship appear in the parasitic representation of both nets?

- Coupling capacitance is a mutual connection between two nets.
- During extraction, the same coupling capacitor is reported in the parasitic data of both nets.
- This allows STA/SI tools to model the interaction from either net's perspective.
- Tools avoid double-counting internally and recognize it as a shared coupling element.

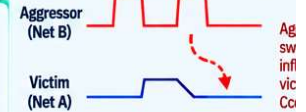
25

How does coupling capacitance affect STA and signal integrity?

Coupling capacitance affects both timing (STA) and signal integrity (SI) by creating interaction between neighboring nets, which can change delays and cause noise.

1 Increased Effective Capacitance

- The switching activity on a neighboring net can make the coupling capacitance appear as additional load.
- This increases the effective capacitance seen by the driver, leading to larger RC delay.
- In STA, this is modeled using coupling-aware delay calculation (e.g., Ceff).



2 Crosstalk

- Coupling capacitance creates electrical interaction between adjacent nets.
- Switching on one net (aggressor) can induce a voltage change on the other net (victim).
- This is called crosstalk.

3 Aggressor / Victim Relationships

- Aggressor: the net that is switching and causes noise.
- Victim: the net that receives the induced noise.
- The effect depends on the relative signal direction (both in same direction → more coupling, opposite direction → less).

4 Delay Impact

- Coupling increases the effective load capacitance, which increases transition delay.
- In worst cases, it can cause setup/hold violations if not properly accounted for.
- STA tools use coupling-aware analysis (e.g., CCS/ECSM, AOCV/POCV with crosstalk).

5 Noise Impact (Signal Integrity)

- Induces unwanted voltage spikes (glitches) on the victim net.
- Can cause false switching, functional errors, or increased bit error rate (BER).
- Critical for high-speed and low-noise (e.g., memory interfaces, SerDes).

6 Why SI-aware timing requires coupling information

- Traditional (lumped) load capacitance is not enough to capture crosstalk effects.
- Modern STA/SI analysis uses coupling capacitance to model real switching scenarios.
- Needed for accurate delay, noise, glitch, and signal integrity analysis.
- Used in signoff flows (e.g., PrimeTime SI, Tempus SI) for timing, noise and power analysis.

Used in Modern Signoff

- STA (Timing)
- SI (Noise / Crosstalk)
- Power Analysis
- Overall Signoff



Key Takeaway:

"A coupling-capacitance entry (e.g., 5 *9:1 *10:1 0.0621) defines a mutual capacitance between two nodes, which affects both **timing (delay)** and **signal integrity (crosstalk)**, and must be considered for accurate signoff."



- Include Coupling in STA/SI
- Analyze SI (Noise / Crosstalk)
- Essential for Modern Signoff

26

What is the *RES section?

The *RES section lists the resistances between nodes of a net, modeling the interconnect (wires and vias) as a distributed RC network.

1 What is *RES?

- The *RES section lists the resistances between nodes of a net extracted from the layout.
- Each entry represents a resistance between two nodes.
- It models the interconnect as a distributed RC network instead of a single lumped resistance.
- The resistances can be from wire segments or vias.

2 Syntax and Example

Syntax:

```
*RES
<res_id> <node1> <node2> <res_value>
```

Example:

```
*RES
1 *9:1 *9:2 4.18
2 *9:2 *9:3 2.71
3 *9:3 *9:4 5.20
```

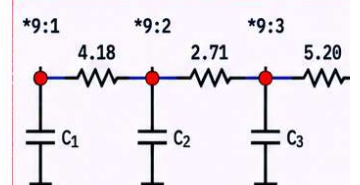
This means there are three resistances

3 Field Description

Field	Description	Example
1	Resistance ID (unique number)	1
*9:1	Start node (first node of the resistor)	*9:1
*9:2	End node (second node of the resistor)	*9:2

4 Visual Representation

Distributed Resistor Network (RC Ladder)



- Used by STA tools to compute accurate delays and signal propagation.

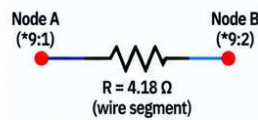
This means there are three resistances connecting node *9:1 to *9:2, *9:2 to *9:3, and *9:3 to *9:4 with the given resistance values (in the units specified in the SPEF header).

Resistance value (in R_UNIT, typically ohms) 4.18

Each resistor represents the resistance of a wire segment between two adjacent nodes

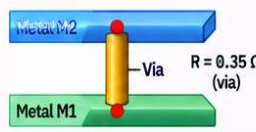
5 Wire vs Via Resistance

Wire-Segment Resistance



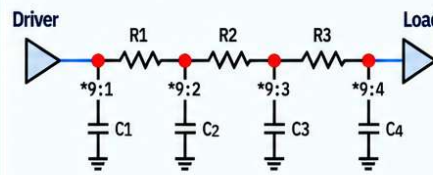
- Models the resistance of a metal wire between two nodes.
- Value depends on metal layer, length, width, and process technology.
- Typically larger for long and thin wires.

Via Resistance



- Models the resistance of a via (connection between metal layers).
- Usually smaller than long wire resistance.
- Important in multi-layer routing.

6 Complete RC Network View



The combination of *RES (resistances) and *CAP (capacitances) forms a distributed RC ladder network that accurately models the real interconnect.

7 Key Takeaways

- *RES lists the resistances between nodes.
- Each entry has: ID, start node, end node, a resistance value.
- Models wire-segment resistance and via resistance.
- Used to build a distributed RC ladder (not lumped R).
- Works together with *CAP to model the full network.
- Helps STA tools compute accurate delay, signal integrity.



Key Message:

"The *RES section provides the interconnect resistances between nodes, modeling the real wire and via parasitics as a distributed RC network for accurate timing analysis."



- Models Wire and Via Resistance
- Part of Distributed RC Network
- Essential for Accurate Signal Integrity

26

What is the *RES section ?

The *RES section lists the resistances between nodes of a net, modeling the interconnect (wires and vias) as a distributed RC network.

Resistor model the wire - as a single

1 What is *RES ?

- The *RES section lists the resistances between nodes of a net extracted from the layout.
- Each entry represents a resistance between two nodes.
- It models the interconnect as a distributed RC network instead of a single lumped resistance.
- The resistances can be from wire segments or vias.
- Used by STA tools to compute accurate delays and signal propagation.

2 Syntax and Example

Syntax:

*RES
<res_id> <node1> <node2> <res_value>

Example:

```
*RES
1 *9:1 *9:2 4.18
2 *9:2 *9:3 2.71
3 *9:3 *9:4 5.20
```

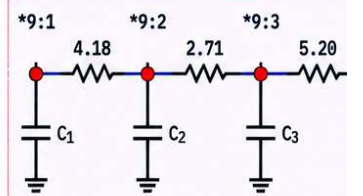
This means there are three resistances connecting node *9:1 to *9:2, *9:2 to *9:3, and *9:3 to *9:4 with the given resistance values (in the units specified in the SPEF header).

3 Field Description

Field	Description	Example
1	Resistance ID (unique number)	1
*9:1	Start node (first node of the resistor)	*9:1
*9:2	End node (second node of the resistor)	*9:2
4.18	Resistance value (in R_UNIT, typically ohms)	4.18

4 Visual Representation

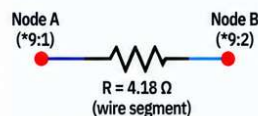
Distributed Resistor Network (RC Ladder)



Each resistor represents the resistance of a wire segment between two adjacent nodes

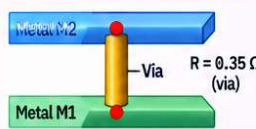
5 Wire vs Via Resistance

Wire-Segment Resistance



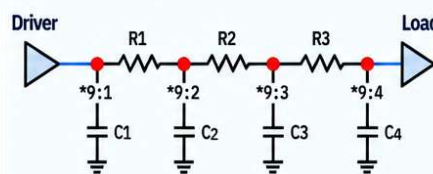
- Models the resistance of a metal wire between two nodes.
- Value depends on metal layer, length, width, and process technology.
- Typically larger for long and thin wires.

Via Resistance



- Models the resistance of a via (connection between metal layers).
- Usually smaller than long wire resistance.
- Important in multi-layer routing.

6 Complete RC Network View



The combination of *RES (resistances) and *CAP (capacitances) forms a distributed RC ladder network that accurately models the real interconnect.

7 Key Takeaways

- *RES lists the resistances between nodes.
- Each entry has: ID, start node, end node, a resistance value.
- Models wire-segment resistance and via resistance.
- Used to build a distributed RC ladder (not lumped R).
- Works together with *CAP to model the full network.
- Helps STA tools compute accurate delay, signal integrity.



Key Message:

"The *RES section provides the interconnect resistances between nodes, modeling the real wire and via parasitics as a distributed RC network for accurate timing analysis."



- Models Wire and Via Resistance
- Part of Distributed RC Network
- Essential for Accurate Signal Integrity

27

How does SPEF represent a distributed RC network ?

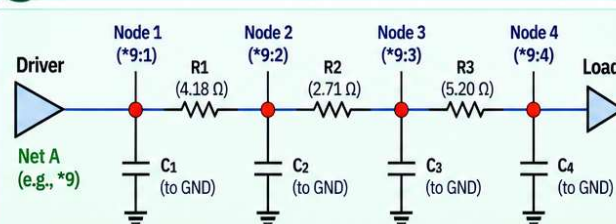
SPEF models an interconnect as a set of resistors and capacitors at multiple nodes, forming a distributed RC network instead of a single lumped RC.

Real world are distributed - not a single R or C

1 Concept Overview

- SPEF divides a wire into multiple RC nodes.
- Each segment has a resistance (*RES) between nodes.
- Each node has ground capacitance (*CAP to ground).
- Nodes can also have coupling capacitances (*CAP between nodes of adjacent nets).

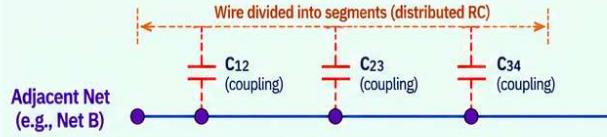
2 Distributed RC Network – Visual Representation



3 Corresponding SPEF Entries (Simplified)

*D_NET (total capacitance for the net)				
*D_NET	*9	0.3847	← total capacitance (sum of all c)	
*RES (segment resistances between nodes)				
*RES				
1	*9:1	*9:2	4.18	
2	*9:2	*9:3	2.71	
3	*9:3	*9:4	5.20	
*CAP (ground and coupling capacitances)				
				Couplin

- This creates a distributed RC network that models the real interconnect more accurately.
- Used by STA tools to compute accurate delay, slew and crosstalk.



*CAP				
*9:1	0.0234	← C1 to ground		
*9:2	0.0456	← C2 to ground		
*9:3	0.0312	← C3 to ground		
*9:4	0.0187	← C4 to ground		
5 *9:1	*10:1	0.0621	← coupling to adjacent	
6 *9:2	*10:2	0.0315		
7 *9:3	*10:3	0.0284		

appear parasitic both nets (*9 and

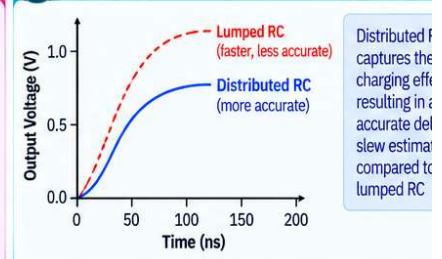
4 RC Nodes and Their Meaning

Item	Description
RC node (*9:1, *9:2, ...)	A point along the net where capacitances are connected.
Segment resistance	Models the resistance of the wire (or via) between two nodes.
Ground capacitance	Capacitance from the node to ground/substrate (modeling wire-to-ground, via-to-substrate, device diffusion, etc.).
Coupling capacitance	Capacitance between this node and a node of a neighboring net.

5 Why Distributed RC is More Accurate ?

- Real interconnect has resistance and capacitance spread along its length, not just at one point.
- Captures Elmore delay more accurately.
- Models location-dependent effects (near driver vs near load).
- Includes coupling effects for crosstalk analysis.
- Better accuracy for long wires, high-speed designs and advanced nodes.
- A single lumped R and C can miss delay, slew and crosstalk effects, especially for long or heavily coupled nets.

6 Lumped vs Distributed RC – Delay Comparison



Key Takeaway:

"SPEF represents a distributed RC network using multiple nodes, resistors (*RES) and capacitors (*CAP), which provides a much more accurate model of real interconnect for timing and signal-integrity analysis."

- Accurate Delay and Slew
- Includes Coupling (Crosstalk)
- Essential for STA and SI Analysis

28

How do SPEF units and naming rules affect interpretation ?

SPEF files define the units for time, capacitance, resistance and inductance, and also specify naming rules using dividers and delimiters. These settings control how you interpret the numerical values and hierarchical names in the file.

Correct units naming rule critical for accurate timing and SI analysis

1 SPEF Unit Definitions

SPEF specifies the units for time, capacitance, resistance and inductance using the following keywords.

Keyword	Meaning	Common Units	Example from SPEF	Notes
*T_UNIT	Time unit	NS (nanosecond) PS (picosecond)	*T_UNIT 1 NS *T_UNIT 1 PS	Used for delay, timing information
*C_UNIT	Capacitance unit	FF (femtofarad)	*C_UNIT 1 FF	All capacitance values (*CAP section) are in this unit
*R_UNIT	Resistance unit	OHM (ohm)	*R_UNIT 1 OHM	All resistance values (*RES section) are in this unit
*L_UNIT	Inductance unit	HENRY (henry)	*L_UNIT 1 HENRY	Used when inductance is extracted (e.g., for SI analysis)

2 Unit Examples and Interpretation

The unit setting defines the scale of all numerical values in the SPEF file.

SPEF Line	Interpretation	Example Value	Actual Value (SI)
*T_UNIT 1 NS	Time unit = 1 ns	10.5	10.5 ns
*T_UNIT 1 PS	Time unit = 1 ps	250	250 ps = 0.25 ns
*C_UNIT 1 FF	Capacitance unit = 1 fF	0.0234	0.0234 fF
*R_UNIT 1 OHM	Resistance unit = 1 ohm	4.18	4.18 Ω
*L_UNIT 1 HENRY	Inductance unit = 1 H	0.001	0.001 H

Always check the unit definitions at the top of the SPEF file before interpreting any numerical value. The same number can represent very different values depending on the unit.

3 Naming Rules in SPEF

SPEF uses specific separators to define hierarchy, pin/net names and bus naming.

Keyword	Meaning	Example	Explanation
*DIVIDER	Hierarchy separator (between instance names)	*DIVIDER /	Separates hierarchical levels. Example: top/u1/u2
*DELIMITER	Pin/Net delimiter (between instance and pin)	*DELIMITER :	Separates instance name and pin name. Example: u1:ZN
*BUS_DELIMITER	Bus naming delimiter (for bus indices)	*BUS_DELIMITER []	Defines how bus bits are written. Example: data[7]

4 Naming Examples

How the naming rules appear in real net and instance names.

Item	Example with Default Settings	Explanation
Hierarchical instance	top/u1/u2	Uses "/" as hierarchy separator (from *DIVIDER /)
Instance pin	u1:ZN	":" separates instance and pin (from *DELIMITER :)
Bus net	data[7]	"[]" used for bus index (from *BUS_DELIMITER [])
Hierarchical bus pin	top/u1:ADDR[3]	Hierarchy + pin delimiter + bus index
Another example	chip/block1/u2:D[15]	Common format in real designs

5 Why Units and Naming Rules Matter ?

- Correct units ensure accurate delay, slew, and capacitance/resistance interpretation in STA tools.
- Wrong unit assumptions (e.g., PS vs NS) can lead to 1000x error in timing.
- Consistent naming rules are essential for tools to correctly parse hierarchical names, pins and buses.
- Helps match nets between SPEF, netlist, SDC and other signoff files.
- Important for correlation across tools (Synopsys, Cadence, etc.) and for multi-corner/multi-mode analysis.

6 Key Takeaways

- *T_UNIT → Time (NS/PS)
- *C_UNIT → Capacitance (FF)
- *R_UNIT → Resistance (OHM)
- *L_UNIT → Inductance (HENRY)
- *DIVIDER → Hierarchy separator (e.g., /)
- *DELIMITER → Instance/Pin separator (e.g., :)
- *BUS_DELIMITER → Bus naming (e.g., [])
- Always verify units and naming rules before SPEF data in STA or SI analysis.

Key Message:

"SPEF units and naming rules define how to read and interpret the data. Getting these right is essential for accurate timing, signal integrity and signoff correlation."

- Check Units
- Understand Naming Rules
- Avoid Interpretation Errors

29

How does SPEF represent different RC corners and min/typ/max values ?

SPEF can be generated for different RC extraction corners (best, typical, worst). Each SPEF file contains parasitic values (R, C, etc.) for that corner, and STA uses the appropriate SPEF along with the timing corner for analysis.

Different SPEF files for different RC corners = accurate signoff

1 What are RC corners in SPEF ?

- Parasitic values (R, C) vary with process, voltage, temperature (PVT), and extraction settings.
- To capture this variation, parasitic extraction is run for multiple corners, generating separate SPEF files.
- Common corners are:

RC_WORST → Maximum parasitics (higher R, higher C)

2 How are min/typ/max represented ?

There are two common ways:

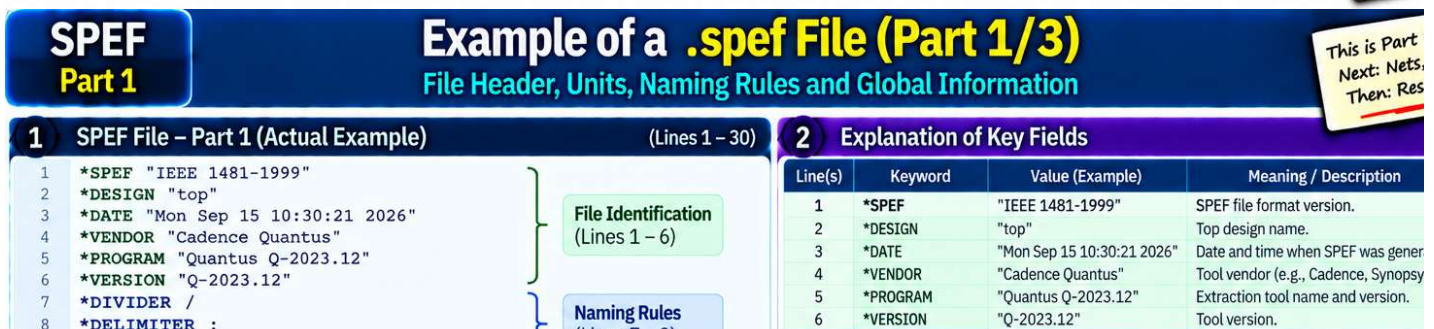
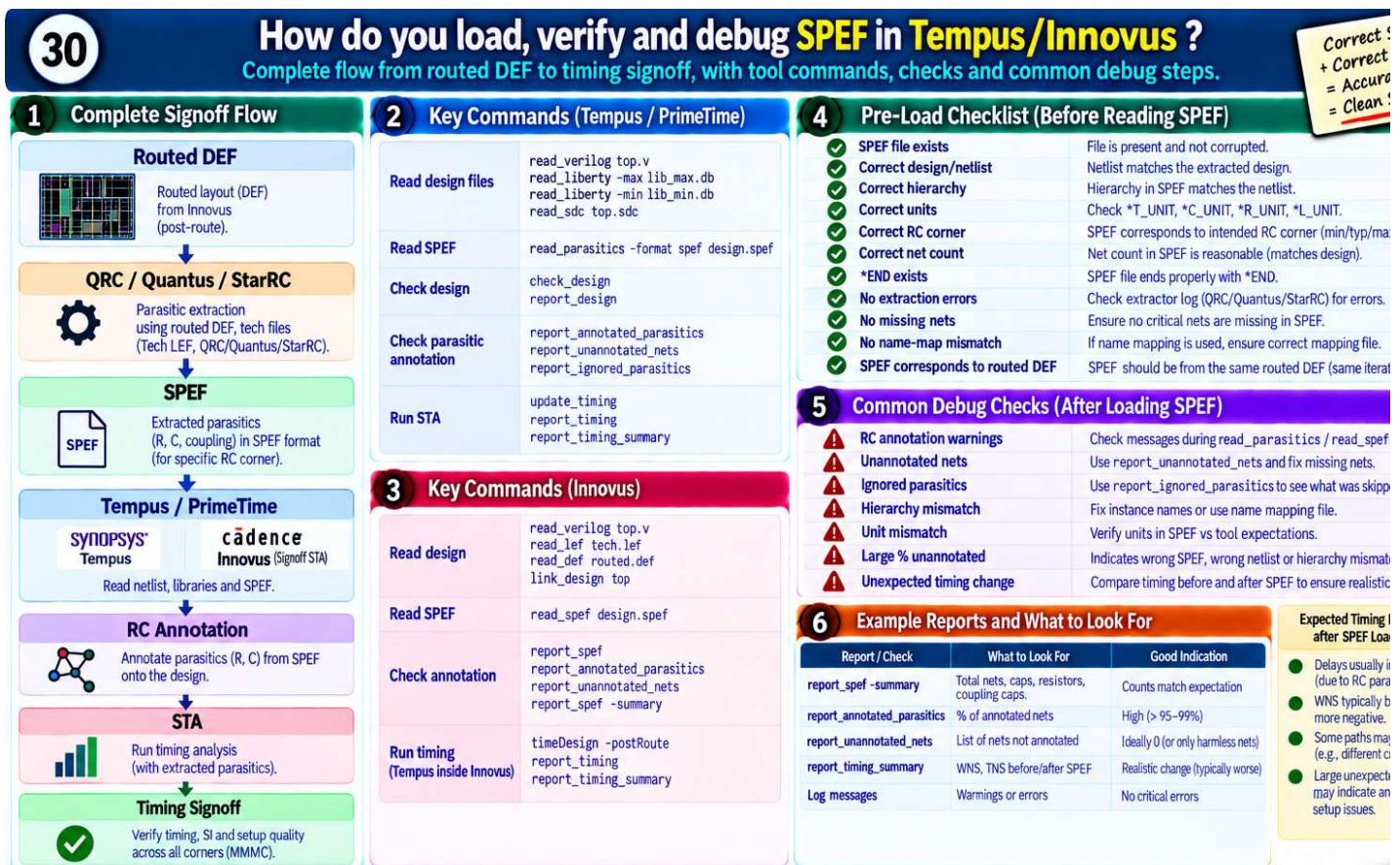
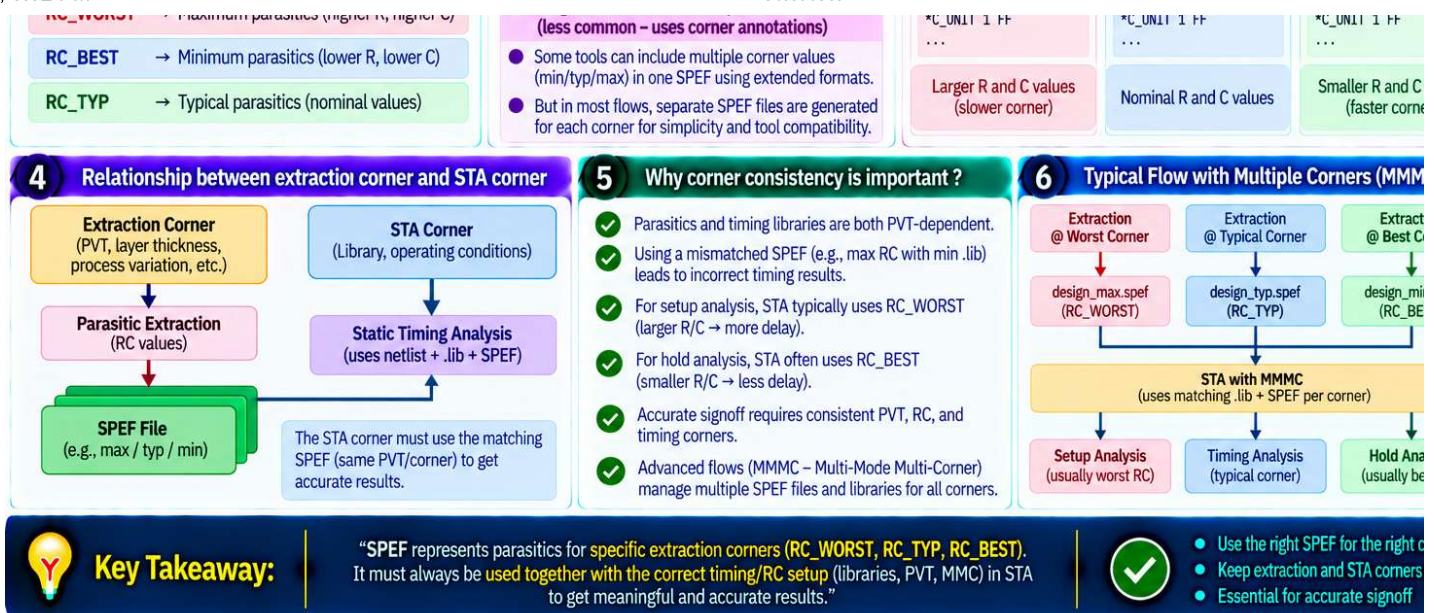
a) Separate SPEF files per corner (most common)

- One SPEF file for RC_WORST (max)
- One SPEF file for RC_TYP (typical)
- One SPEF file for RC_BEST (min)

b) Single SPEF file with multiple corners

3 Example: Different SPEF files per corner

RC_WORST (MAX)	RC_TYP	RC_BEST (MIN)
design_max.spef	design_typ.spef	design_min.spef
*SPEF "IEEE 1481-1999"	*SPEF "IEEE 1481-1999"	*SPEF "IEEE 1481-1999"
*DESIGN "top"	*DESIGN "top"	*DESIGN "top"
*DATE "..."	*DATE "..."	*DATE "..."
*R_UNIT 1 OHM	*R_UNIT 1 OHM	*R_UNIT 1 OHM
*C_UNIT 1 FF	*C_UNIT 1 FF	*C_UNIT 1 FF



9	*BUS_DELIMITER []	
10	*T_UNIT 1 NS	
11	*C_UNIT 1 FF	
12	*R_UNIT 1 OHM	
13	*L_UNIT 1 HENRY	
14	*D_UNIT 1 MICRON	
15	*A_UNIT 1 SQUARE_MICRON	
16	*V_UNIT 1 VOLT	
17	*TEMP 25	
18	*PROCESS "typical"	
19	*CORNER "RC_TYP"	
20	*MODE "extracted"	
21	*DESIGN_FLOW "signoff"	
22	*TOTAL_NETS 10234	
23	*TOTAL_INSTANCES 5421	
24	*TOTAL_PINS 18976	
25	*TOTAL_CAP 3.82145e+04	
26	*TOTAL_RES 1.92453e+04	
27	*TOTAL_COUPLING_CAP 7.36211e+03	
28	*EXTRACTION_TOOL "Quantus"	
29	*EXTRACTION_DATE "Mon Sep 15 10:28:11 2026"	
30	*COMMENT "Example SPEF file - Part 1 (Header and Units)"	

Unit Definitions
(Lines 10 – 16)Process / Corner Info
(Lines 17 – 21)Statistics
(Lines 22 – 30)

7	*DIVIDER	/	Hierarchy separator between instan
8	*DELIMITER	:	Separator between instance and pin
9	*BUS_DELIMITER	[]	Defines bus naming (e.g., data[7]).
10	*T_UNIT	1 NS	Time unit (used for delay informatio
11	*C_UNIT	1 FF	Capacitance unit (1 femtofarad).
12	*R_UNIT	1 OHM	Resistance unit.
13	*L_UNIT	1 HENRY	Inductance unit (optional, for RLC).
14	*D_UNIT	1 MICRON	Distance unit.
15	*A_UNIT	1 SQUARE_MICRON	Area unit.
16	*V_UNIT	1 VOLT	Voltage unit.
17	*TEMP	25	Extraction temperature (°C).
18	*PROCESS	"typical"	Process condition.
19	*CORNER	"RC_TYP"	RC corner (typ/min/max).
20	*MODE	"extracted"	Extraction mode.
21	*DESIGN_FLOW	"signoff"	Flow stage (e.g., signoff, ECO).
22-27	*TOTAL_*	Various	Statistics about nets, instances, pins, res, etc.
28-29	*EXTRACTION_*	Various	Tool and extraction timestamp.
30	*COMMENT	"..."	User comment (optional).

3 Key Takeaways from Part 1

- ✓ The header defines the design, tool, units and naming rules used in the SPEF file.
- ✓ Units are very important – always check T_UNIT, C_UNIT, R_UNIT, etc., before interpreting values.
- ✓ Naming rules (DIVIDER, DELIMITER, BUS_DELIMITER) must match your design/netlist.
- ✓ Corner and process information tells you for which PVT/RC condition the SPEF was generated.
- ✓ Statistics give a quick sanity check (number of nets, total caps/res, etc.).

4 What's Next ?

In Part 2, we will see:

- 1 *D_NET section (nets, pins, total capacitance)
- 2 *CONN section (instance connections)
- 3 *CAP section (ground and coupling capacitances)

This is only the header part.
The actual parasitic data (nets, caps, resistances) comes in the next parts.

SPEF Part 2/3

Example of a .spef File (Part 2/3)

*D_NET Section – Nets, Pins and Total Capacitance

This part of the SPEF file contains the parasitic information for each net, including its connected pins, total capacitance and a reference to the capacitors and resistors.

1 SPEF File – Part 2 (D_NET Section)

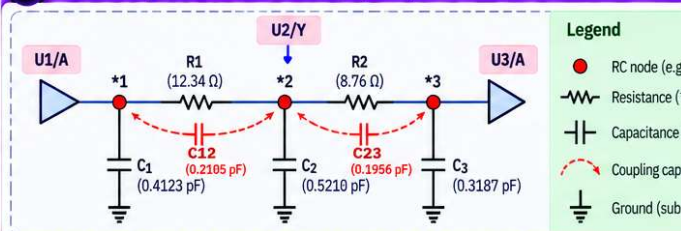
(Example: Lines 31 – 70)

31	*D_NET net1 3.4821e-12	# Net name and total capacitance (F)
32	*CONN	
33	PIN U1/A	# Connected pins/ports for this net
34	PIN U2/Y	
35	PIN U3/A	
36	*CAP	
37	1 *1 0.4123e-12	# Node capacitances (to GND or substrate)
38	2 *2 0.5210e-12	
39	3 *3 0.3187e-12	
40	4 *1 *2 0.2105e-12	# Coupling capacitances (between nodes)
41	5 *2 *3 0.1956e-12	
42	*RES	
43	1 *1 *2 12.34	# Segment resistances (ohm)
44	2 *2 *3 8.76	
45	*END	# End of net
46		
47	*D_NET clk 8.7215e-12	# Another net (e.g., clock net)
48	*CONN	
49	PIN U10/CLK	# Pins connected to this net
50	PIN U15/CLK	
51	PIN U20/CLK	
52	*CAP	
53	1 *4 1.2103e-12	# Ground capacitances
54	2 *5 1.0345e-12	
55	3 *6 0.9987e-12	
56	4 *4 *5 0.4321e-12	# Coupling capacitances
57	5 *5 *6 0.3892e-12	
58	*RES	
59	1 *4 *5 5.67	# Segment resistances
60	2 *5 *6 6.21	
61	*END	# End of net

2 Explanation of the D_NET Section

Keyword / Entry	Example	Meaning
*D_NET	net1 3.4821e-12	Start of a new net. Net name and total capacitance (sum of all caps and res).
*CONN	PIN U1/A	List of pins/ports connected to this net.
PIN	U1/A	Instance pin connected to the net.
*CAP	1 *1 0.4123e-12	Capacitor entries (node or coupling).
*RES	1 *1 *2 12.34	Resistor entries between nodes.
*END	-	End of the current net.

3 Visual Representation of the Net (net1)



4 Total Capacitance Check

For net1:

- Node caps: $0.4123 + 0.5210 + 0.3187 = 1.2520$ pF
- Coupling caps: $0.2105 + 0.1956 = 0.4061$ pF
- Total = **1.6581 pF** (This may differ slightly from 3.4821e-12 in real cases due to more caps in actual file)

The total capacitance mentioned in *D_NET is the sum of all node and coupling capacitances for that net (across the entire design, including all nodes).

5 Key Takeaways

- ✓ *D_NET section gives parasitics for each net.
- ✓ Lists all connected pins (useful for cross-checking connectivity).
- ✓ Includes both ground and coupling capacitances.
- ✓ Includes segment resistances between RC nodes.
- ✓ Total capacitance should match the sum of all caps for the net.
- ✓ Always verify important nets (e.g., clock, reset, critical paths).

6 Next in Part 3

In Part 3, we will see:

- 1 More *D_NET examples (complex nets with more nodes)
- 2 Remaining sections (e.g., *D_INST, *NAME_MAP, *END)
- 3 Practical tips to verify and debug SPEF files

Nets are the heart of a SPEF file. Always verify critical nets.

SPEF Part 3/3

Example of a .spef File (Part 3/3)

*CONN, *CAP and *RES Sections – Complete Example and Explanation

This part shows the remaining sections of the SPEF file: instance connections, capacitances (ground and coupling) and resistances.

1 SPEF File – Part 3 (CONN, CAP and RES Sections)

(Example: Lines 71 – End)

71	*CONN		# Instance connections in the design
72	U1	net1 A	
73	U1	VDD VPWR	
74	U1	VSS VGND	
75	U2	net1 Y	
76	U2	net2 A	
77	U2	net3 Y	

2 Explanation of Sections

Section	Entries (Example)	Meaning	Key Points
*CONN	U1 net1 A U1 VDD VPWR U2 net1 Y	Instance connections between nets and pins.	• Format: <instance> <net> <pin> • Shows which instances are connected to which nets. • Useful for connectivity checks.
*CAP	1 net1 0.4123e-12	Capacitances of nets.	• Format: <id> <node/net1> [node/net2] • If 2 nodes – coupling cap.

This is Part 3. Now you have a complete example.


```

77 U3 net2 1
78 U3 net3 A
79 U4 net3 Y

80 *CAP                                     # Ground and coupling capacitances
81 1 net1 0.4123e-12                       # Ground cap of net1
82 2 net2 0.5210e-12                       # Ground cap of net2
83 3 net3 0.3187e-12                       # Ground cap of net3
84 4 net1 net2 0.2105e-12                 # Coupling cap between net1 and net2
85 5 net2 net3 0.1956e-12                 # Coupling cap between net2 and net3

86 *RES                                     # Segment resistances
87 1 net1 N1 N2 12.34                     # R between N1 and N2 on net1
88 2 net1 N2 N3 8.76                      # R between N2 and N3 on net1
89 3 net2 N4 N5 5.67                      # R between N4 and N5 on net2
90 4 net3 N6 N7 6.21                      # R between N6 and N7 on net3

91 *END                                     # End of SPEF file
92 *COMMENT "End of example SPEF file - Part 3/3"

```

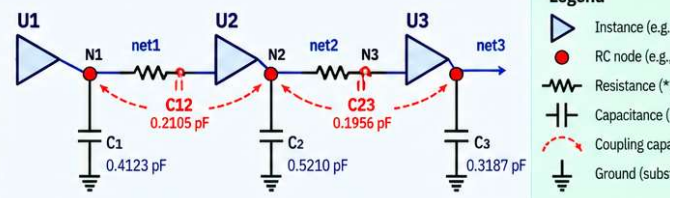
4 Key Takeaways from Part 3

- ✓ *CONN section maps instances to nets and pins.
- ✓ *CAP section includes both ground and coupling capacitances.
- ✓ *RES section models the distributed resistance of nets (wire segments and vias).
- ✓ Node names (e.g., N1, N2) represent RC nodes along the net.
- ✓ Coupling capacitances are between two nets, not to ground.
- ✓ The *END statement must be present.
- ✓ Together with Part 1 (header/units) and Part 2 (nets), this completes a valid SPEF file.

	4 net1 net2 0.2105e-12	- ground capacitance - coupling capacitance	• If 2 nodes → coupling cap • If 1 node → ground cap (to substrate) • Values are in units defined by *C_UNIT (e.g., FF).
*RES	1 net1 N1 N2 12.34 2 net1 N2 N3 8.76	Resistances between RC nodes (segments or vias).	• Format: <id> <net> <from_node> <to_node> • Models distributed resistance of the net • Can include wire segment R and via R.
*END	-	End of SPEF file.	• Must be present.

3 Visual Representation

SPEF represents a distributed RC network with ground and coupling capacitances



5 Final Notes



- The actual SPEF files in real designs can be much larger with thousands of nets, nodes and capacitance/resistance entries.
- Always check the units, naming rules and RC corner information (from Part 1).
- Use STA tools (Tempus / PrimeTime) to read, verify and analyze the SPEF.
- A correct and verified SPEF is essential for accurate timing and signoff.

You are now ready to read and interpret a SPEF file